Lecture 4 or 5
grain boundaries

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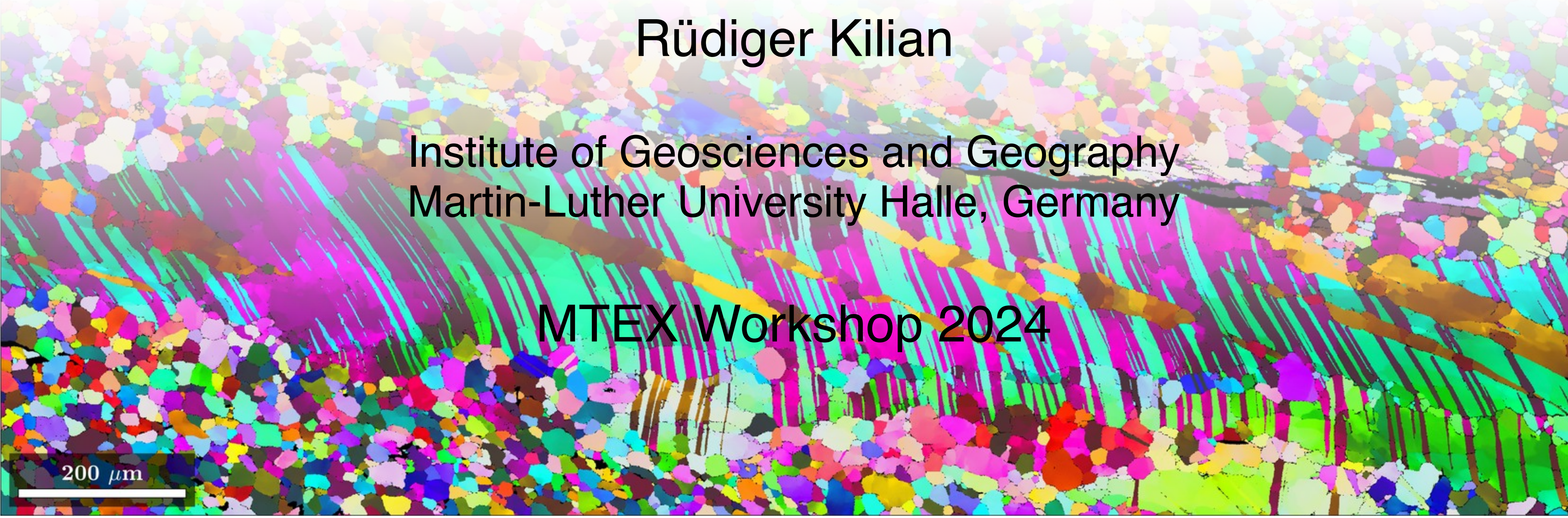


3. Grain boundaries - content

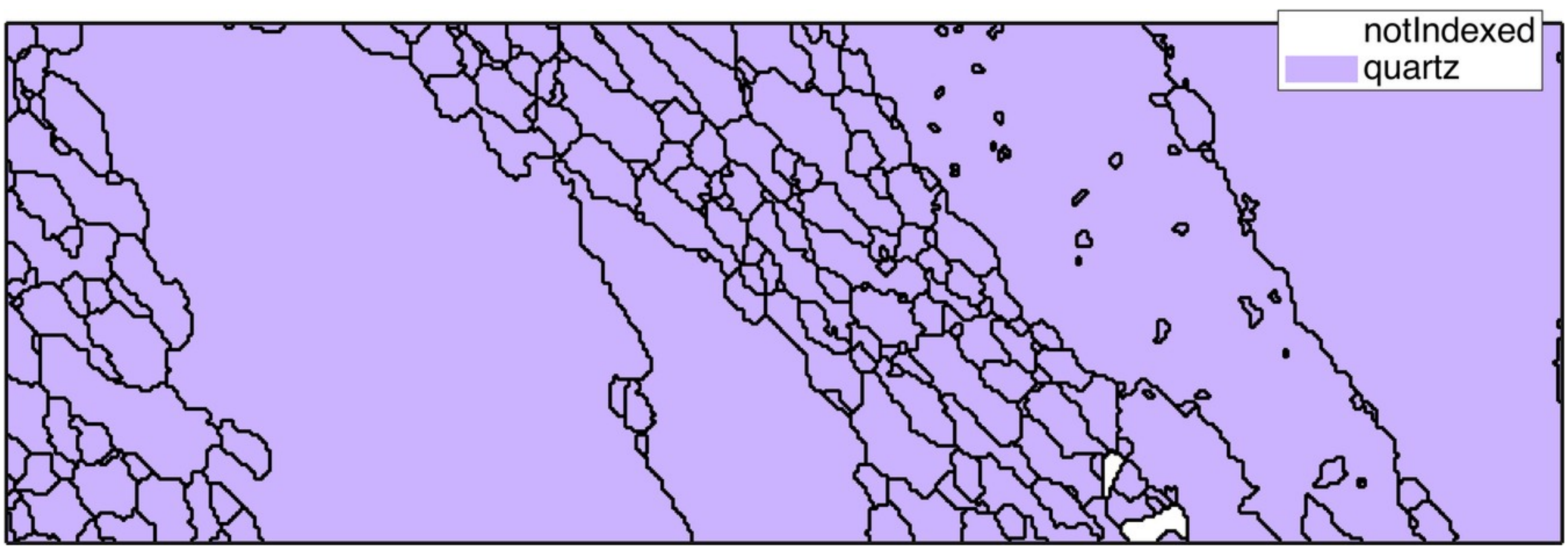
- general information
- relation EBSD – grains – grain boundary
- properties
  - built-in
  - direction
  - (mis)orientation
- selecting grain boundaries based on
  - position
  - misorientations and directions
  - any other prop
- triple points
- merging grains

3. Grain boundaries - general

- general information

```
>> grains
grains = grain2d
 Phase  Grains  Pixels     Mineral  Symmetry
     0       2     164  notIndexed
     1     243   58636      quartz      -3m1

 boundary segments: 7115
 inner boundary segments: 853
 triple points: 357
```

```
% unless 'noBoundary' is given, grains plot
% with boundary
>> plot(grains)
```

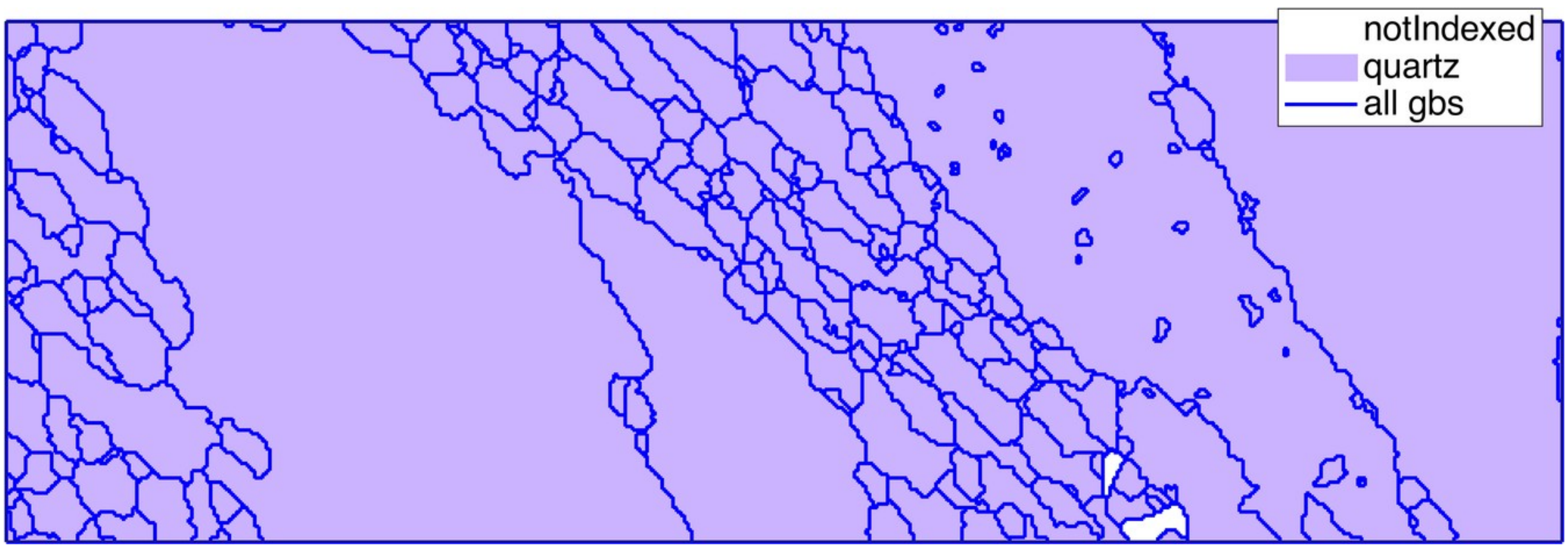


3. Grain boundaries - general

- general information

```
>> grains.boundary

ans = grainBoundary

 Segments   mineral 1   mineral 2
        8  notIndexed  notIndexed
     1202  notIndexed      quartz
     5905      quartz      quartz
```

```
plot(grains)
hold on
plot(grains.boundary, ...
     'linecolor','b', ...
     'displayName','all gbs')
```

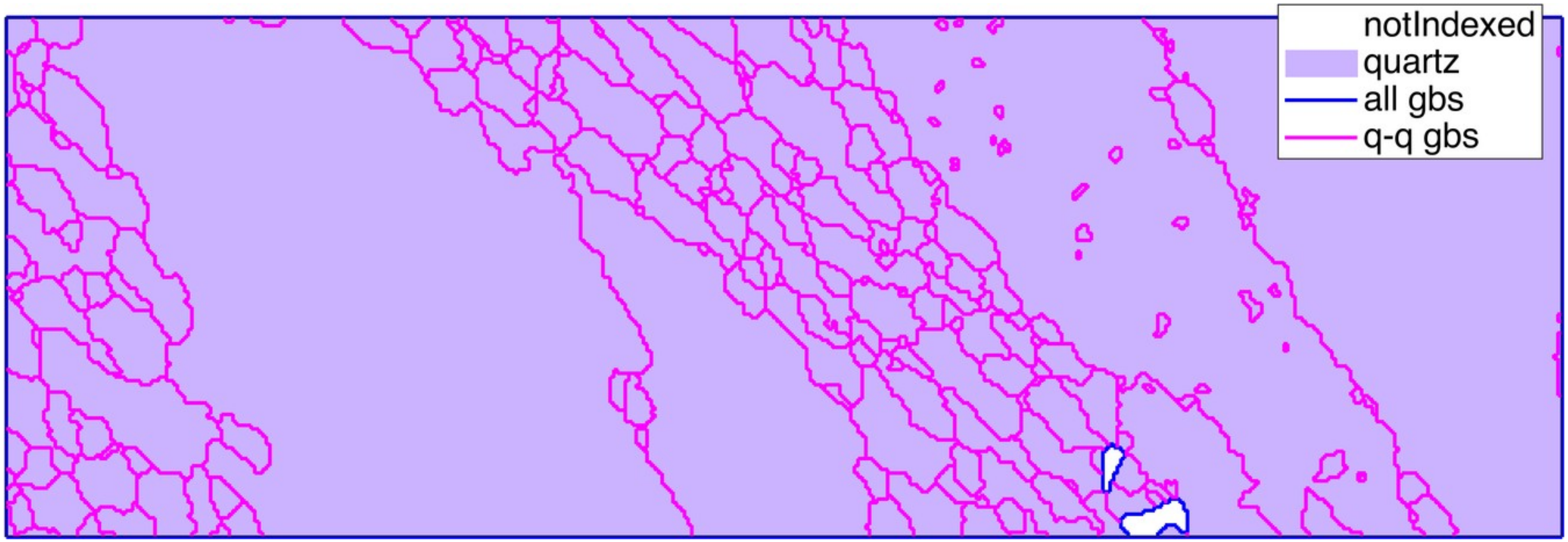


3. Grain boundaries - general

- general information

```
>> grains.boundary('q','q')

ans = grainBoundary

 Segments   mineral 1   mineral 2
     5905      quartz      quartz
```

```
plot(grains)
hold on
plot(grains.boundary, ...
     'linecolor','b', ...
     'displayName','all gbs')
hold on
plot(grains.boundary('q','q'),...
     'linecolor',m',...
     'Fuchsia','displayName','q-q gbs')
```

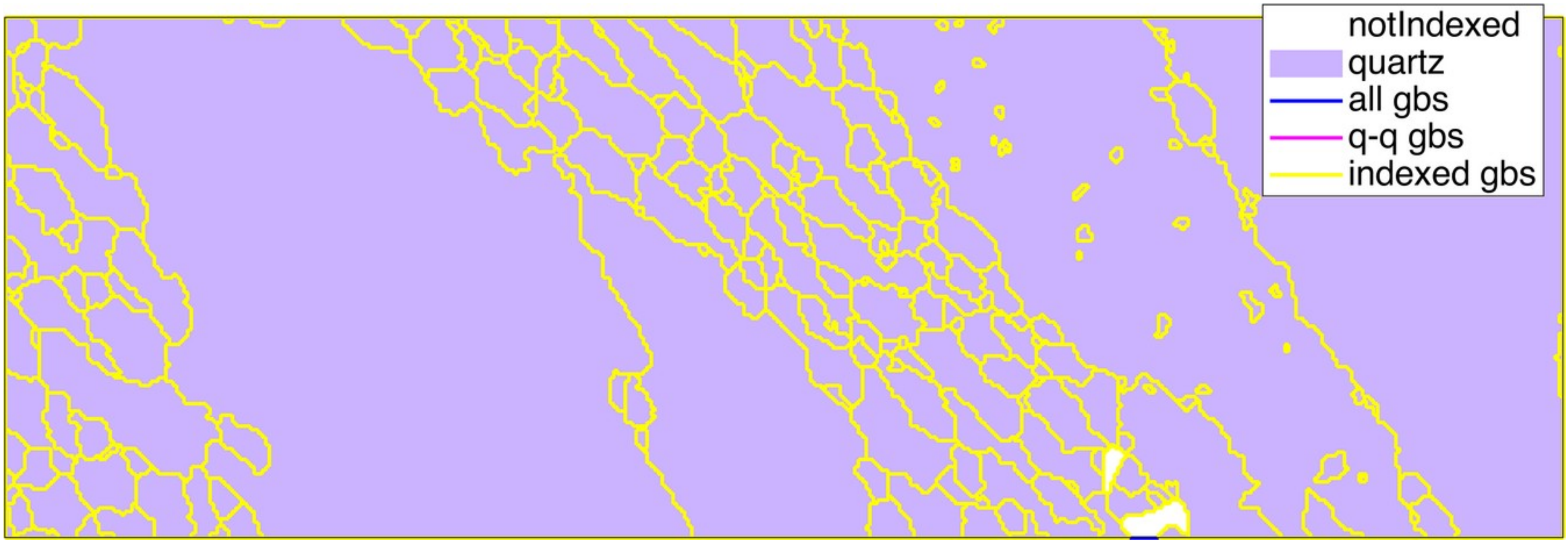


3. Grain boundaries - general

- general information

```
>> grains('indexed').boundary

ans = grainBoundary

 Segments   mineral 1  mineral 2
     1202  notIndexed     quartz
     5905      quartz     quartz
```

```
plot(grains.boundary, ...
     'linecolor','b', ...
     'displayName','all gbs')
hold on
plot(grains.boundary('q','q'),...
     'linecolor','Fuchsia',...
     'displayName','q-q gbs')
hold on
plot(grains('indexed').boundary,...
     'linecolor','y',...
     'displayName','indexed gbs')
```

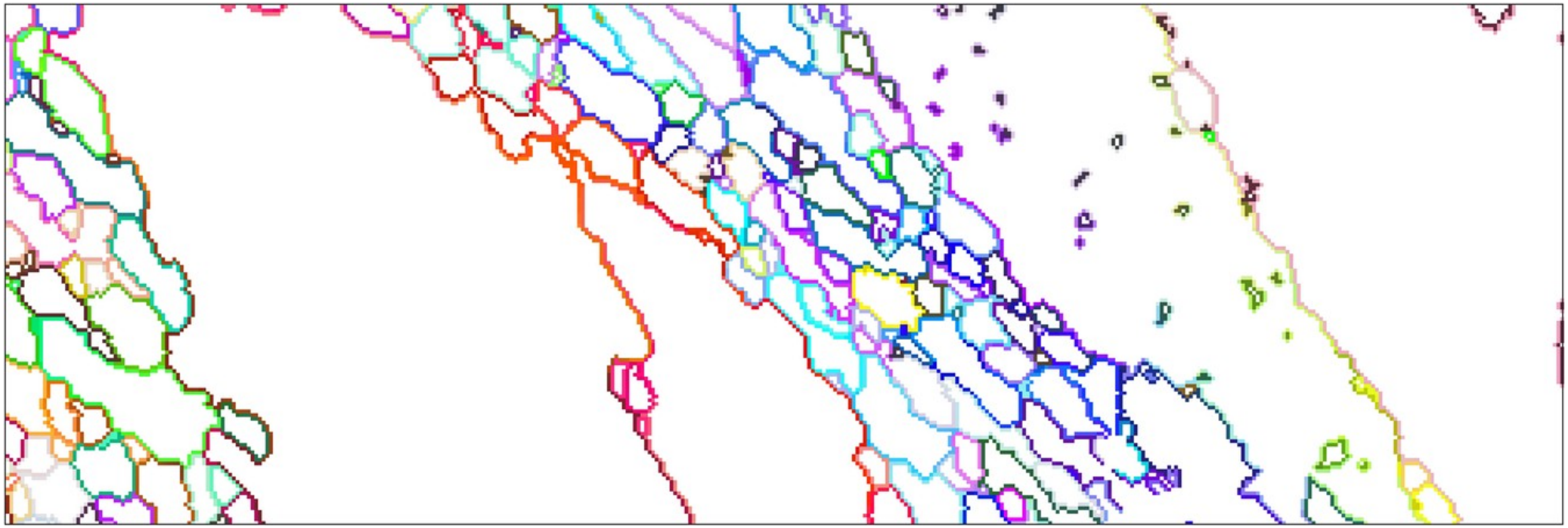


3. Grain boundaries - general

- relation EBSD – grains – grain boundary

```
>> grains.boundary.ebsdId(1:5,:)

ans =

          0        20371

          0        20371

     162286       162285

     162286       163018

          0       327012
```

```
e = ebsd('id',grains.boundary('q','q').ebsdId);

plot(e,e.orientations)
```

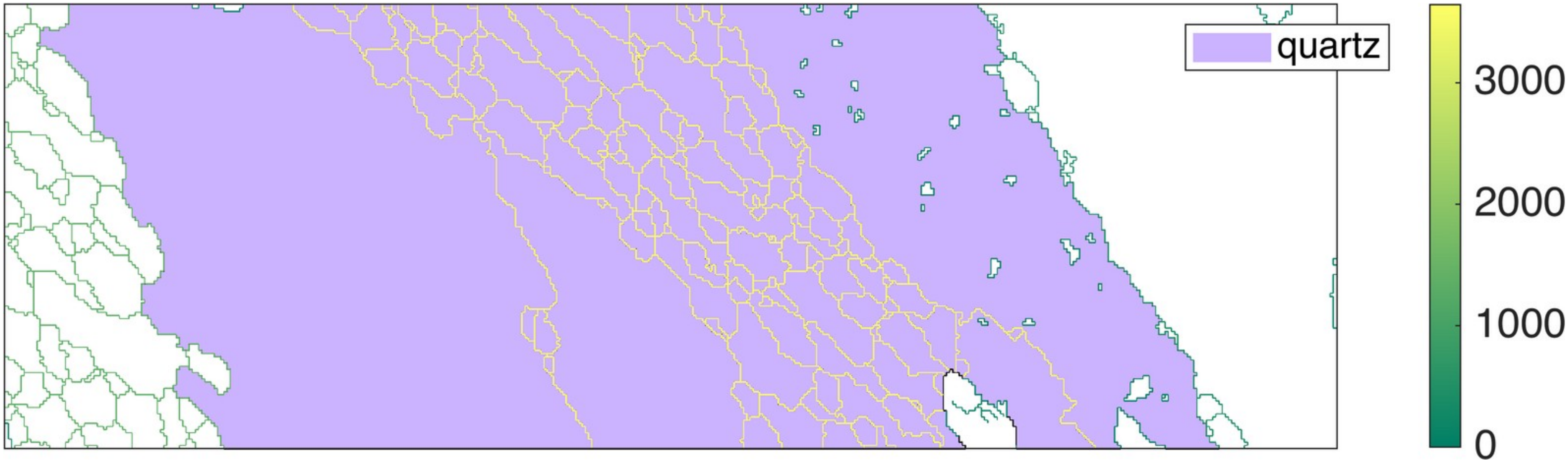


3. Grain boundaries - general

- relation EBSD – grains – grain boundary

```
>> grains.boundary(555:560).grainId

ans =

     0    85

     0    85

    85   127

    85   127

    76    81

    81    85
```

```
gbq = grains.boundary('q','q');
compS = gbq.componentSize;
gbcond = compS==max(compS);

plot(grains('id',unique(gbq(gbcond).grainId)))
hold on
plot(gbq,compS)
hold off
```



3. Grain boundaries - general

- other basic properties and checks

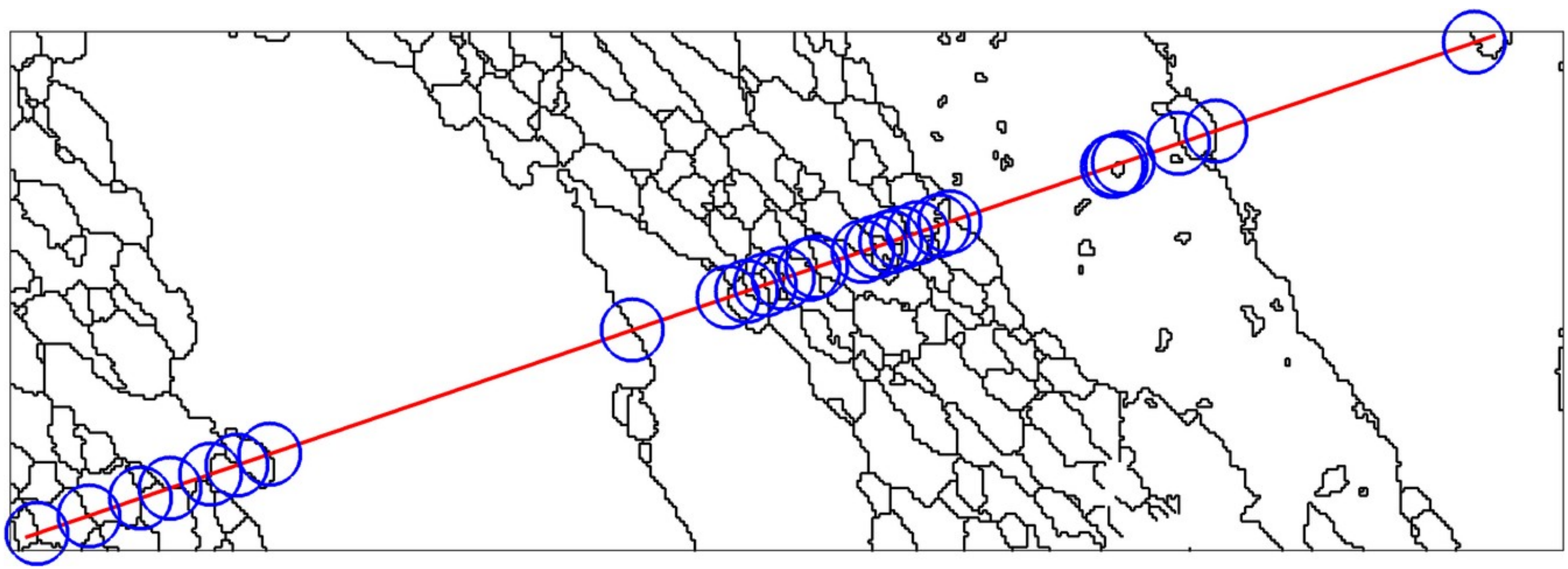
```
>> grains.boundary.hasGrain(grains(123));
>> grains.boundary.hasPhase('q');
>> grains.boundary.isIndexed;
>> grains.boundary.inpolygon([10,10,50,50]);
>> grains.boundary.isTwinning(twinlaw,3*degree);

...
```

ebsdId	id of the EBSD pixel on both sides of the grain boundary (nx2)
phaseId	id of the phase on both sides of the grain boundary (nx2)
grainId	id of the grains on both sides of the grain boundary (nx2)
misorientation	misorientation between orientations at ebsdId(:,1) and ebsdId(:,2)
F	F(:,1), F(:,2) ids of the two vertices V of the segment
segLength	length of the segment
componentId	id of the connected component with respect to all segments
componentSize	number of segments within the component
midPoint	mid point of the segment (vector3d)
direction	vector connecting adjacent vertices (vector3d)
curvature	curvature
triplePoints	triple point list
x,y	unsorted list of x,y coordinates

3. Grain boundaries

- additional functions working on grain boundaries
  - intersect – get line intersects

```
gbq = grains.boundary('q','q');
plot(gbq)
xy=ginput(2);
[xi,yi,segLength] = intersect(gbq,xy(1,:),xy(2,:));
hold on
plot(xy(:,1),xy(:,2),'-r')
hold on
plot(xi(~isnan(xi)),yi(~isnan(yi)),'ob')
hold off
```

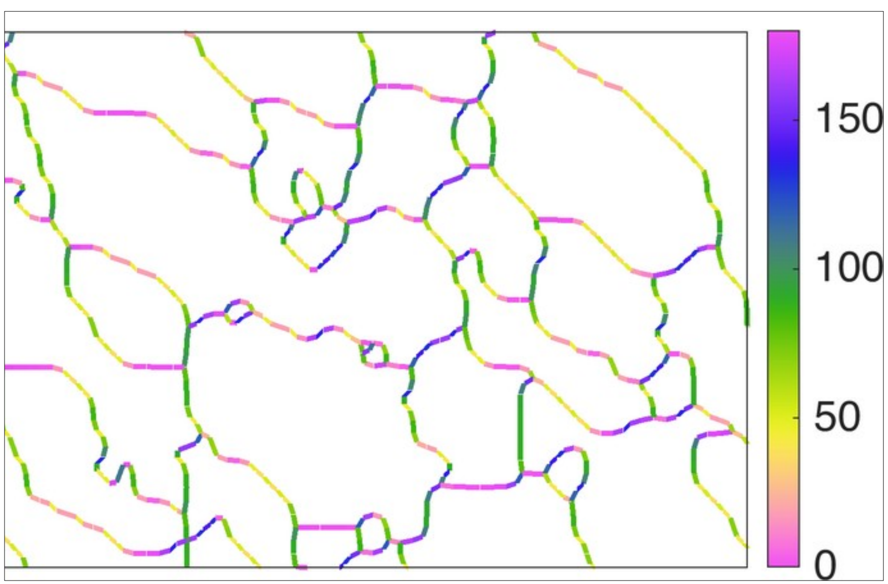


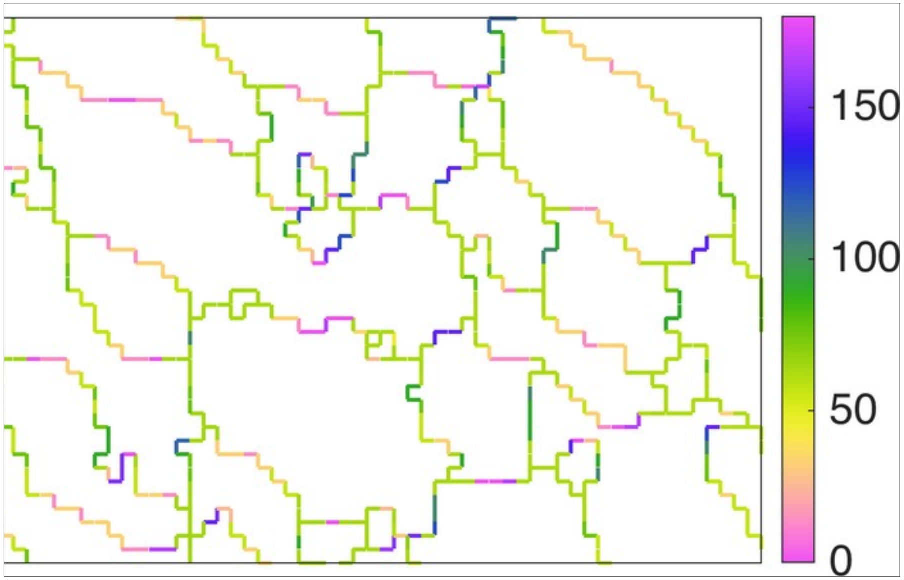
3. Grain boundaries - geometry

- additional functions working on grain boundaries
  - direction and calcMeanDirection

```
gbd = smoothgrains.boundary('q','q').direction;
plot(smoothgrains.boundary('q','q'),mod(gbd.rho,pi)./degree)


md = grains.boundary('q','q').calcMeanDirection(2)
plot(grains.boundary('q','q'),mod(md.rho,pi)./degree)
```

```
>> gbd

gbd = vector3d
 size: 5905 x 1
 antipodal: true
```

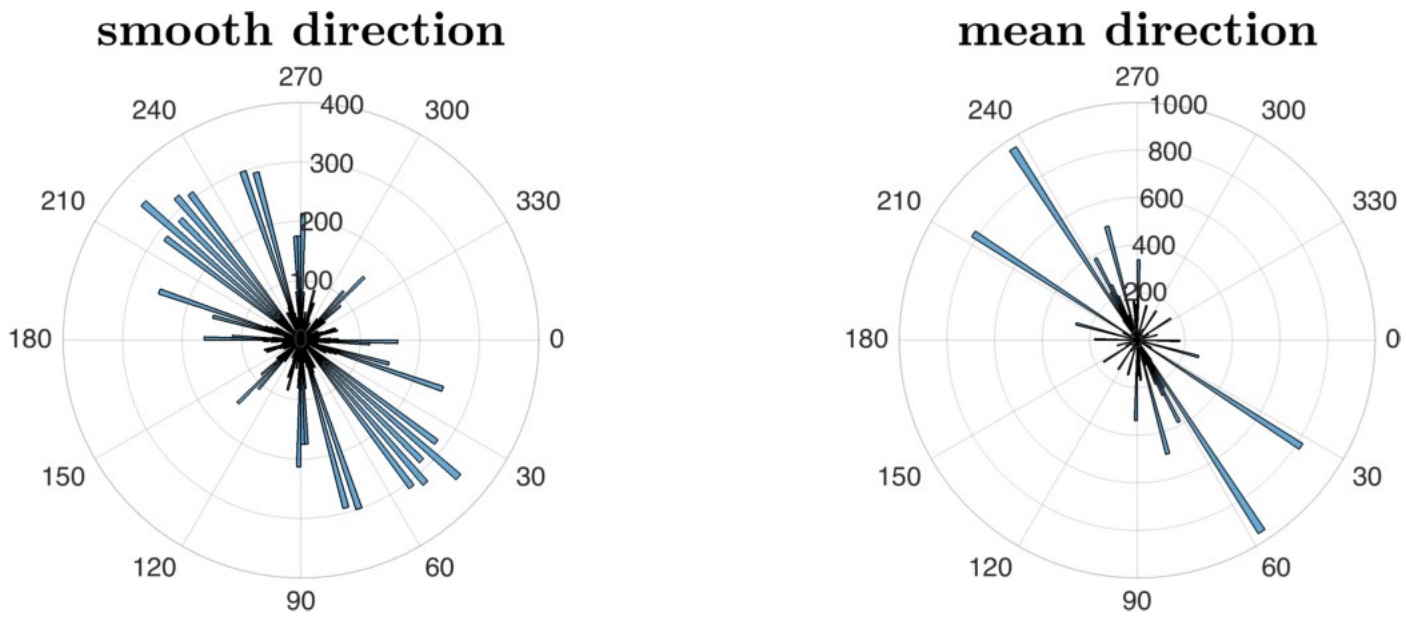




3. Grain boundaries - geometry

- additional functions working on grain boundaries
  - direction and calcMeanDirection

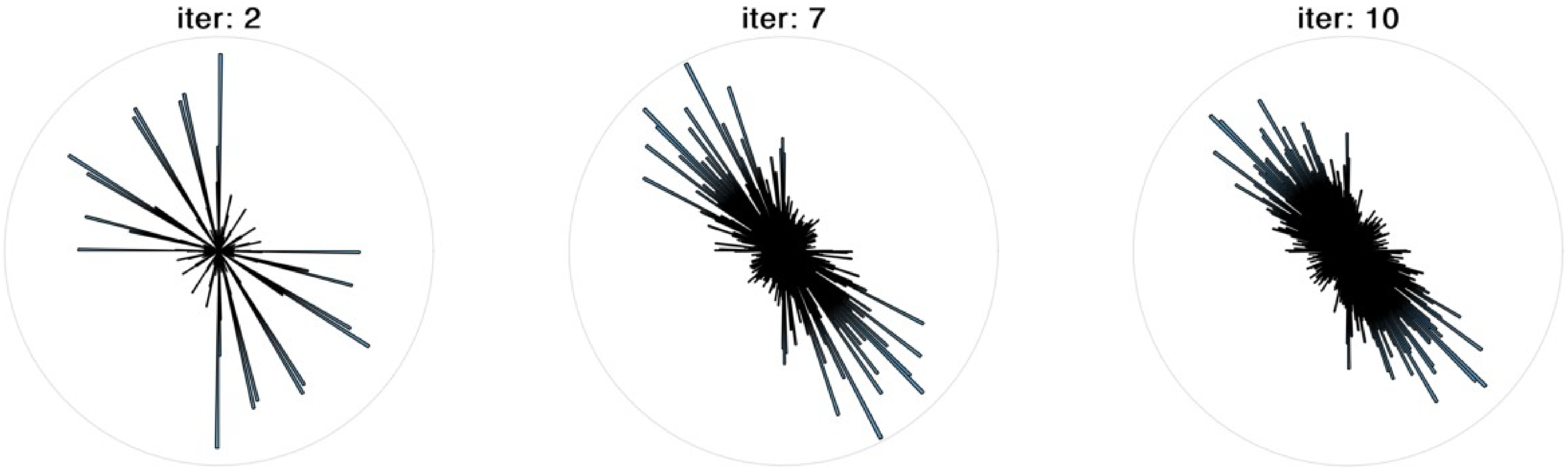
```
histogram(gbd,180)
mtexTitle('smooth direction')

histogram(md,180)
mtexTitle('mean direction')
```



3. Grain boundaries - geometry

- smoothing and grain boundary direction

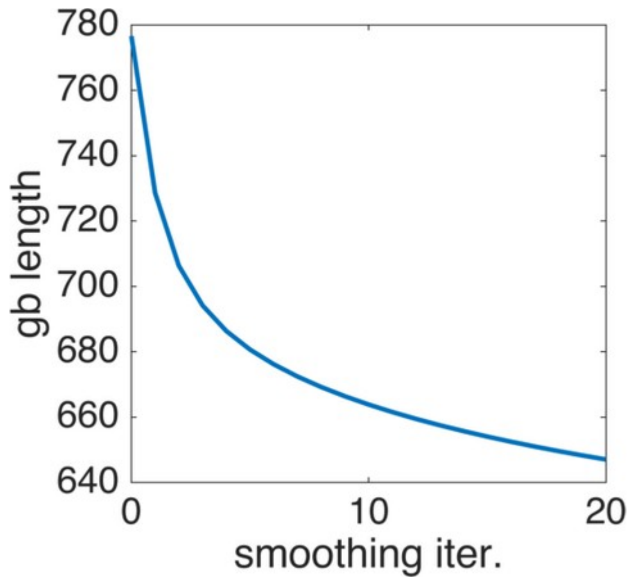
```
sgrains = smooth(grains,n,'moveTriplePoints')
gb = sgrains.boundary('q','q')
histogram(gb.direction);
```

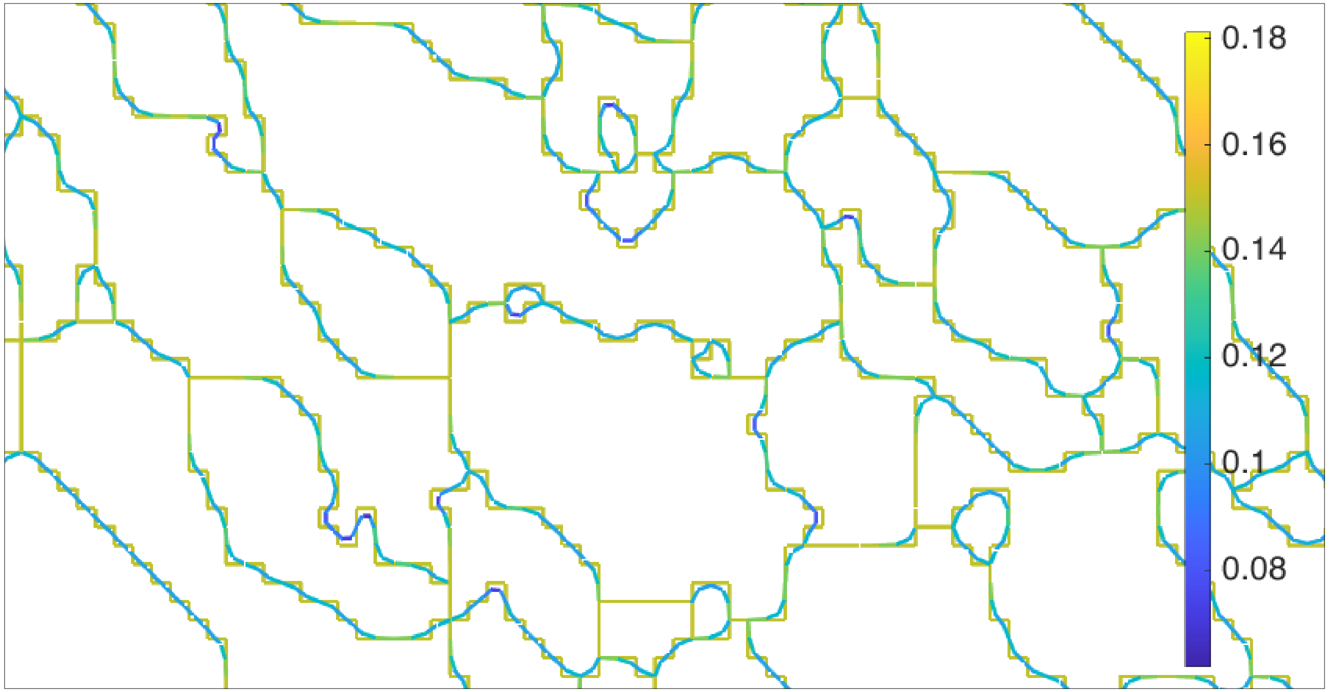


3. Grain boundaries - geometry

- many of these properties only make sense on smooth grains

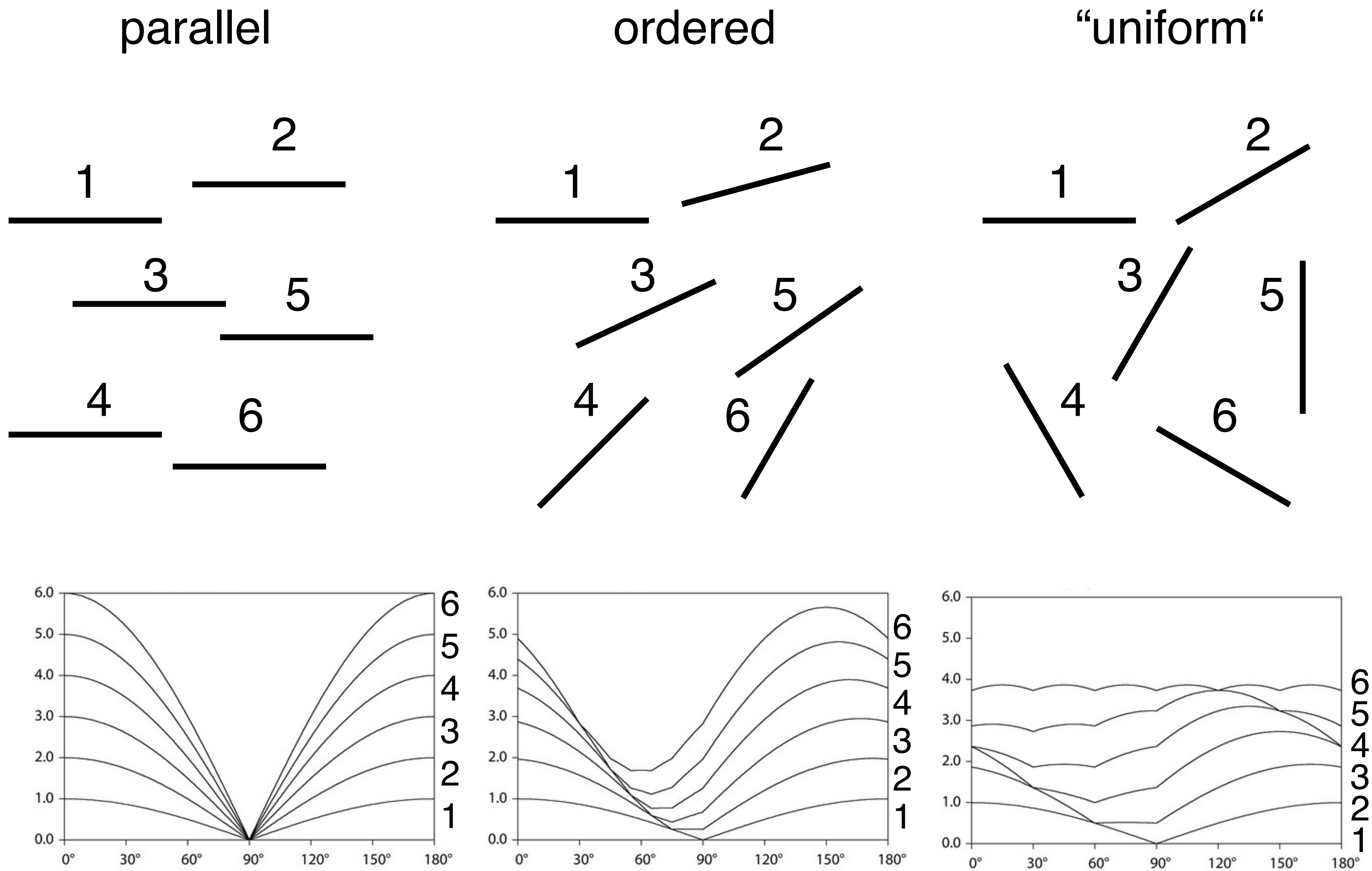
```
iter = 0:20;
sl(1) = sum(grains.boundary('q','q').segLength)
for i = iter
  sgrains = smooth(grains,i+1)
  sl(i+1)=sum(sgrains.boundary('q','q').segLength)
end
plot(iter,sl)
```





3. Grain boundaries - general

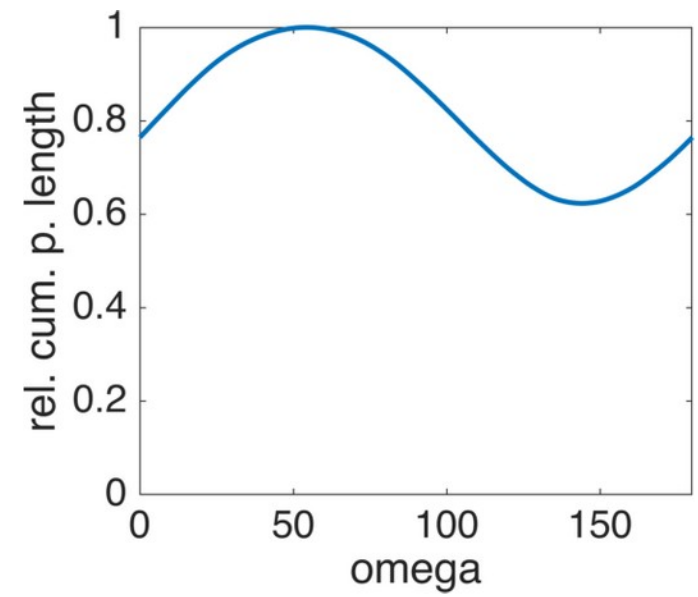
- surfor: cumulative projection function of lines / grain boundary segments

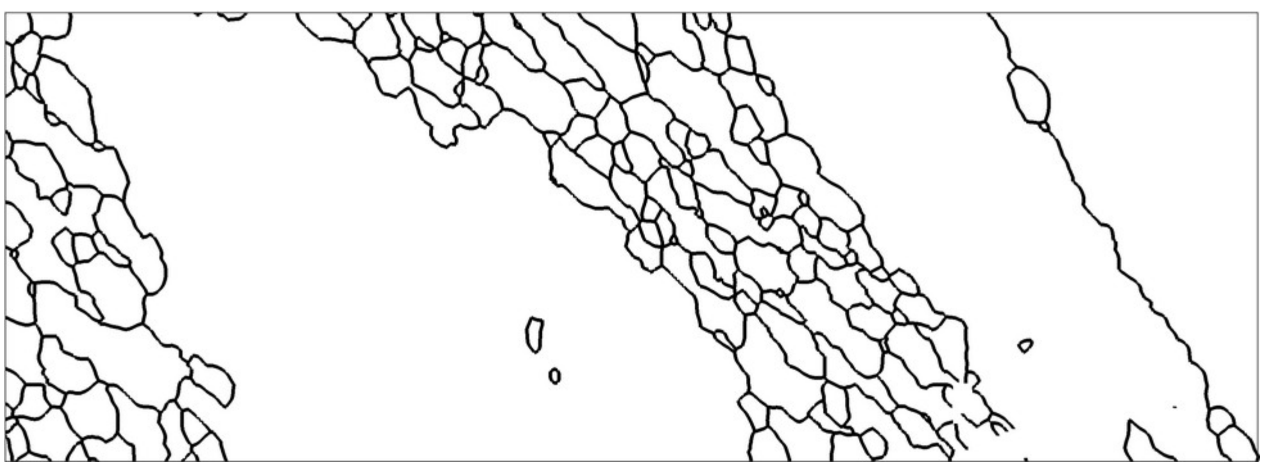


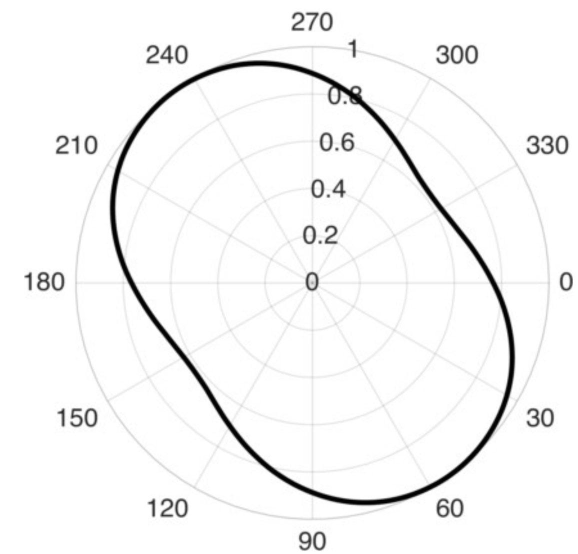
3. Grain boundaries - general

- surfor: cumulative projection function of lines / grain boundary segments

```
omega = [0:360]*degree;
cumpl = surfor(sgrains.boundary('q','q'),omega)
plot(omega,cumpl)

sshape = shape2d.byRhoTheta(cumpl,omega)
plot(sshape,'linewidth',2)
```



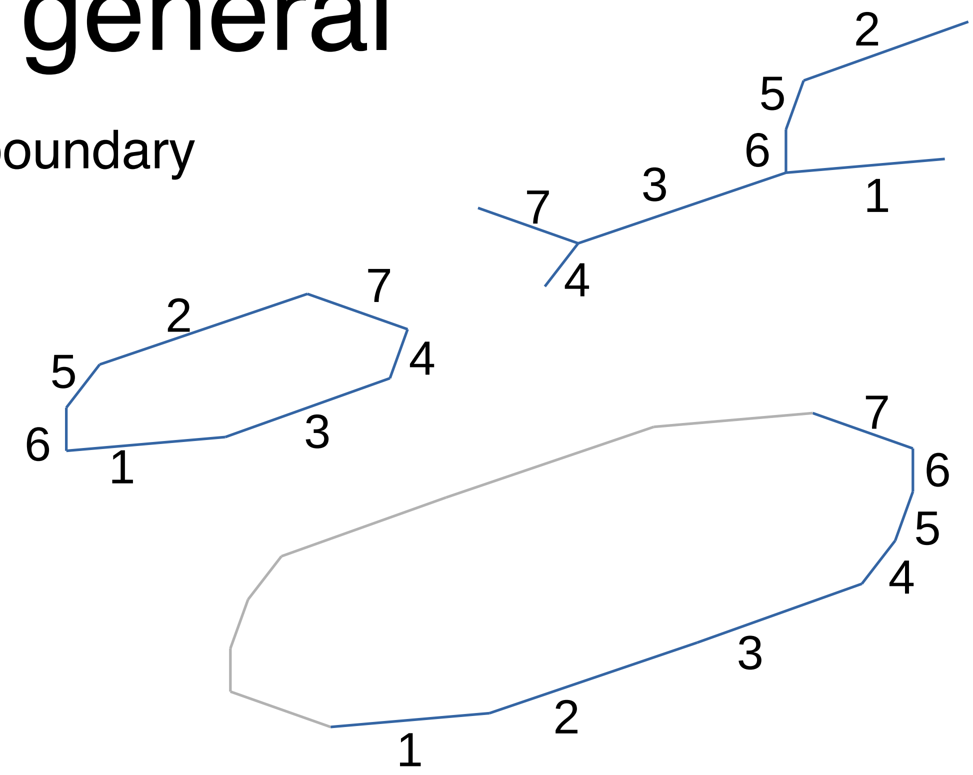




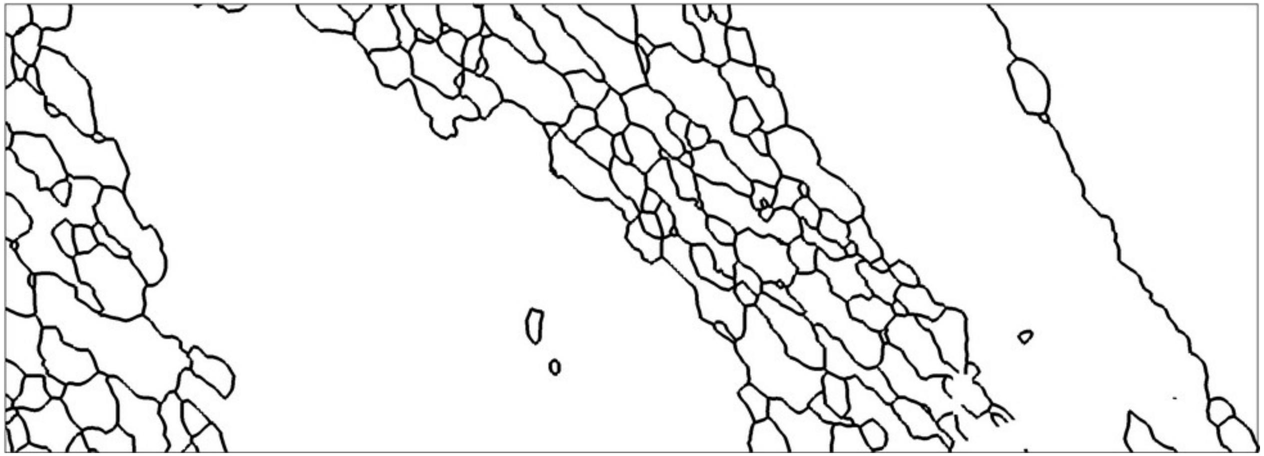
- based on SURFOR by R. Heilbronner (Panozzo,1984)

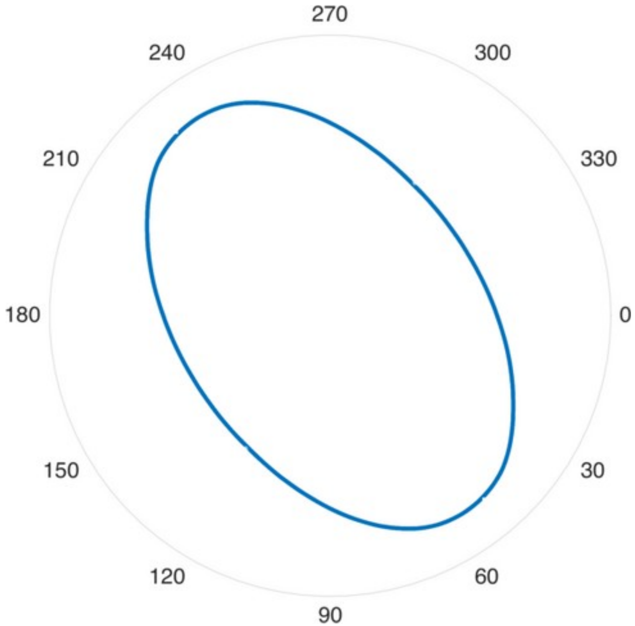
3. Grain boundaries - general

- characteristic shape: cumulative, ordered boundary




```
cShape =
characteristicShape(sgrains.boundary('q','q'))
plot(cShape,'plain','linewidth',2)


angle(cShape.caliper('longest'), ...
      cShape.caliper('shortest'))/degree

» 89.0638
```



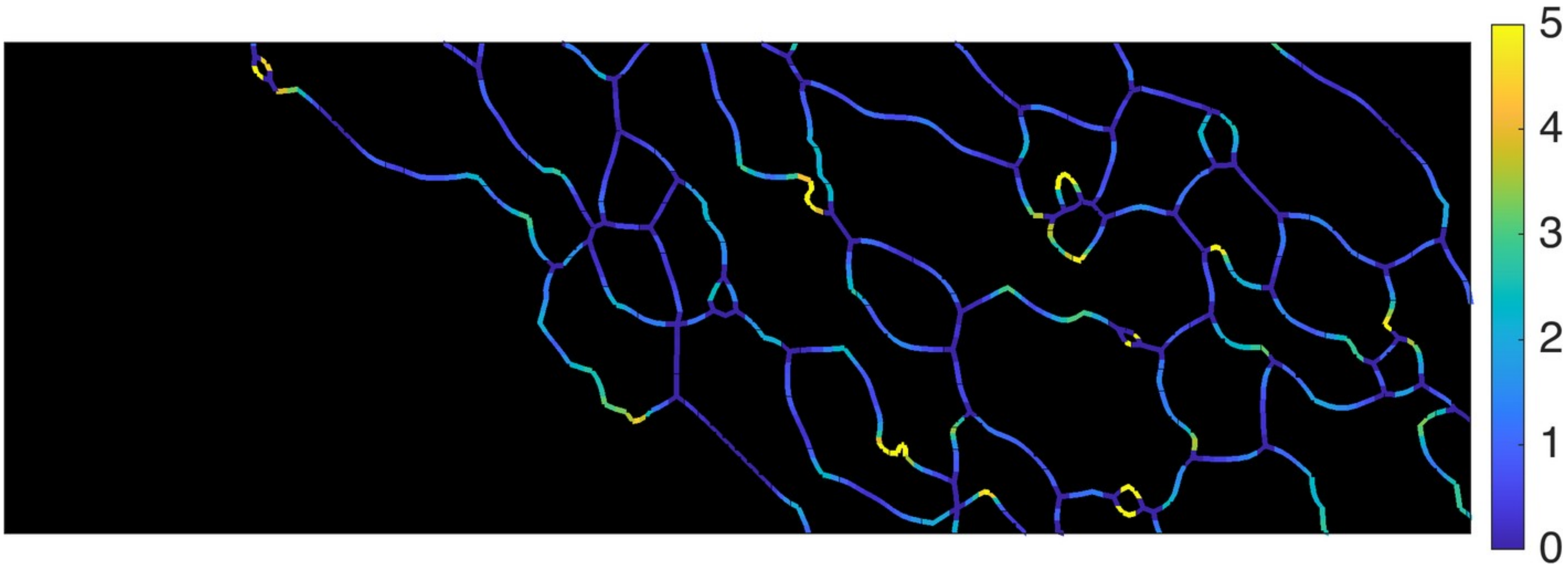


- orthorhomic: strain (?)
- monoclinic: !strain but e.g. dyn. rxx

3. Grain boundaries - general

- curvature (1/R): for all grains, absolute value

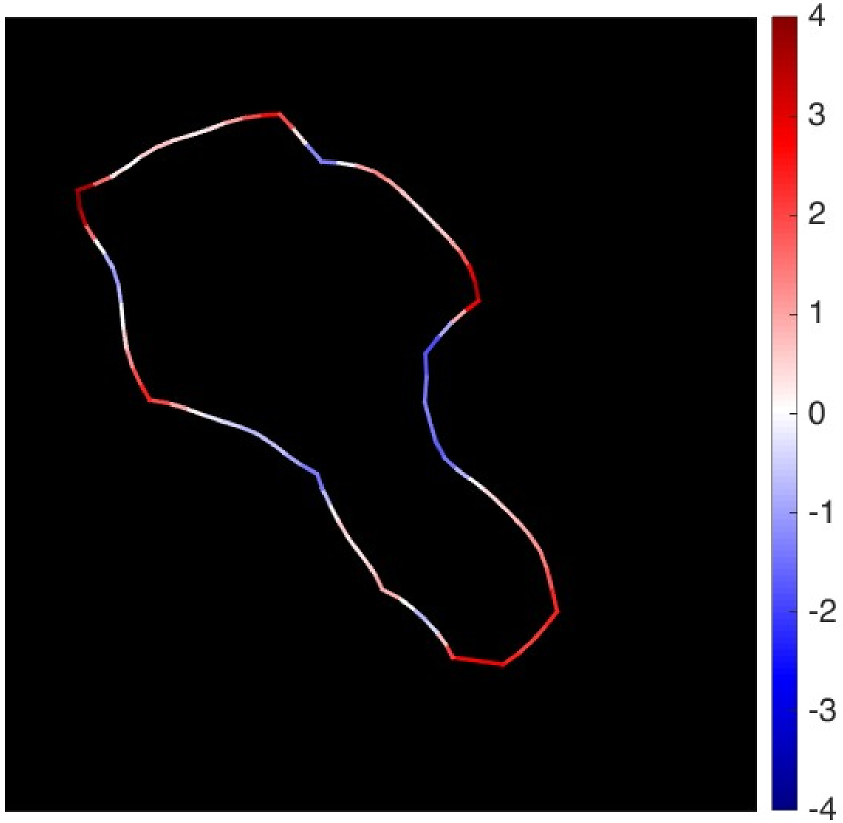
```
gb = sgrains.boundary('q','q');
% all at once
plot(sgrains,'FaceColor','k')
hold on
plot(gb,gb.curvature(2))
hold off
```



3. Grain boundaries - general

- curvature (1/R): for individual grains, signed value

```
gb = sgrains('id',79).boundary('q','q');
% all at once
plot(sgrains,'FaceColor','k')
hold on
plot(gb,gb.curvature(2))
hold off
```

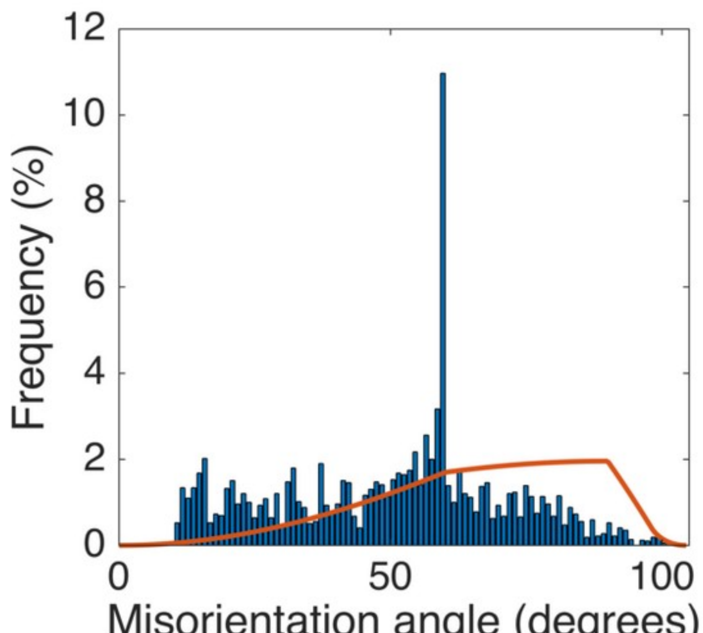


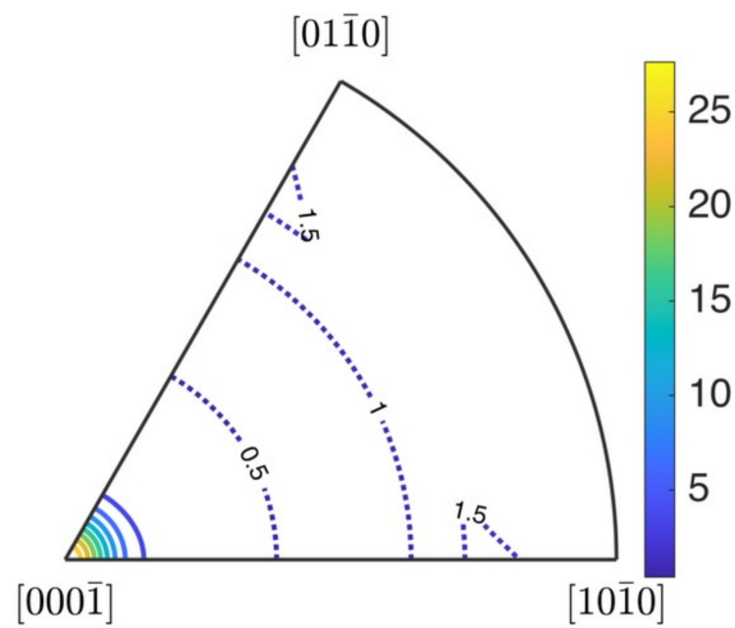
3. Grain boundaries - misorientations

- misorientations angles / axes

```
mori = grains.boundary('q','q').misorientation

>> mori.axis
 ans = Miller (show methods, plot)
 size: 5905 x 1
 mineral: quartz (-3m1, X||a*, Y||b, Z||c*)

>> mori.angle
    1.0236
    0.9607
    1.5829
    ...
```

```
plotAngleDistribution(mori,'resolution',1*degree)
hold on
cs = ebsd('q').CS
plotAngleDistribution(ebsd('q').CS)
hold off

ax = calcAxisDistribution(cs,cs,'antipodal')
plot(ax,'contour','ShowText','on', 'LineStyle',':')
hold on
plotAxisDistribution(mori,'contour')
hold off
```

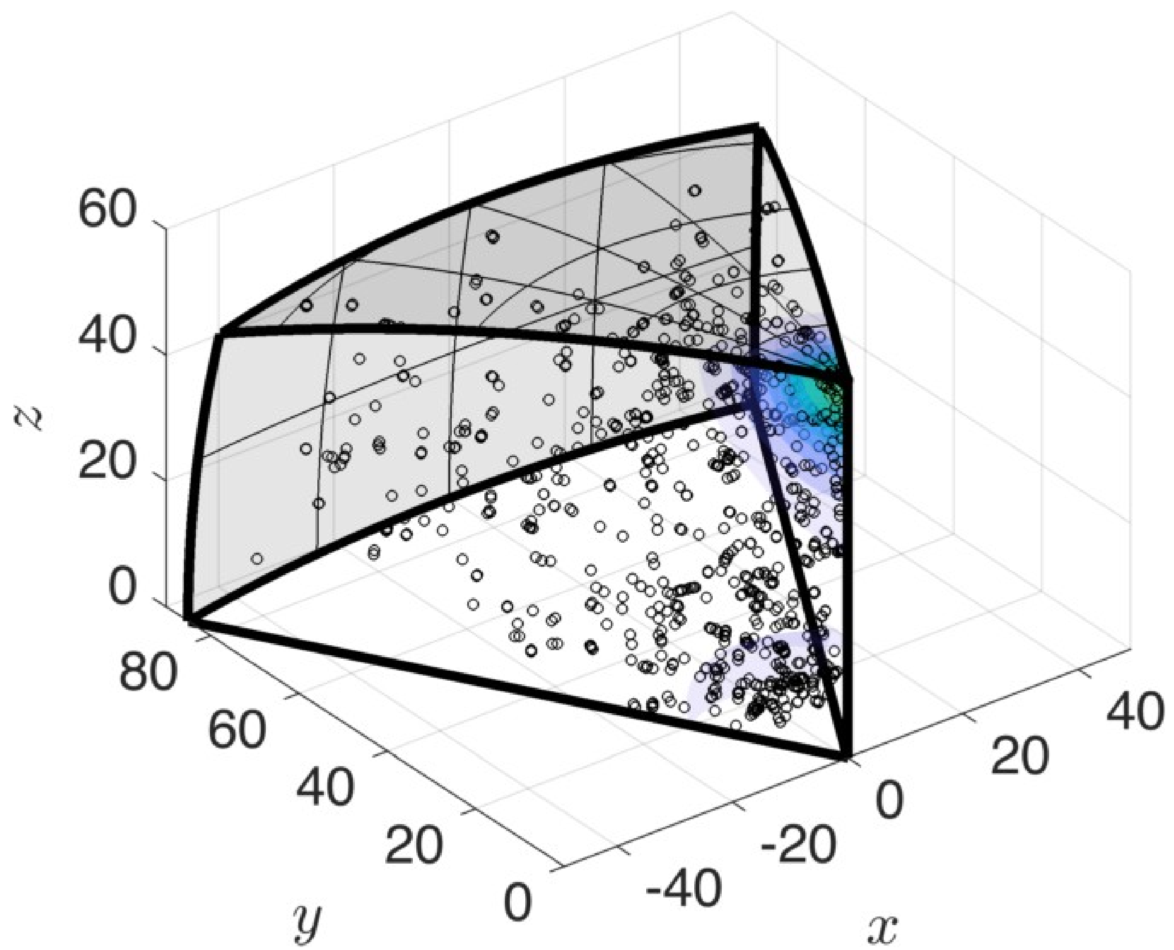




3. Grain boundaries - misorientations

- misorientations axis-angle space

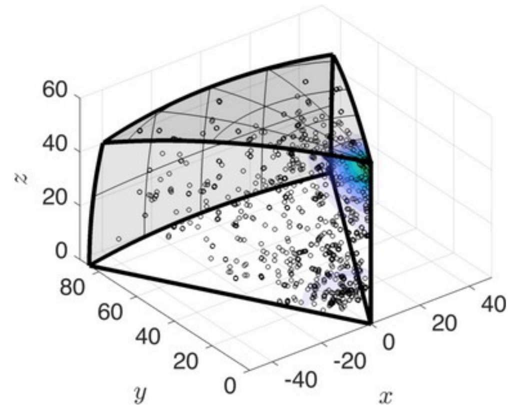
```
mdf = calcDensity(mori,'halfwidth',10*degree)
plot(mdf,'contourf','axisAngle')
hold on
plot(mori,'axisAngle','points',1000)
hold off
```

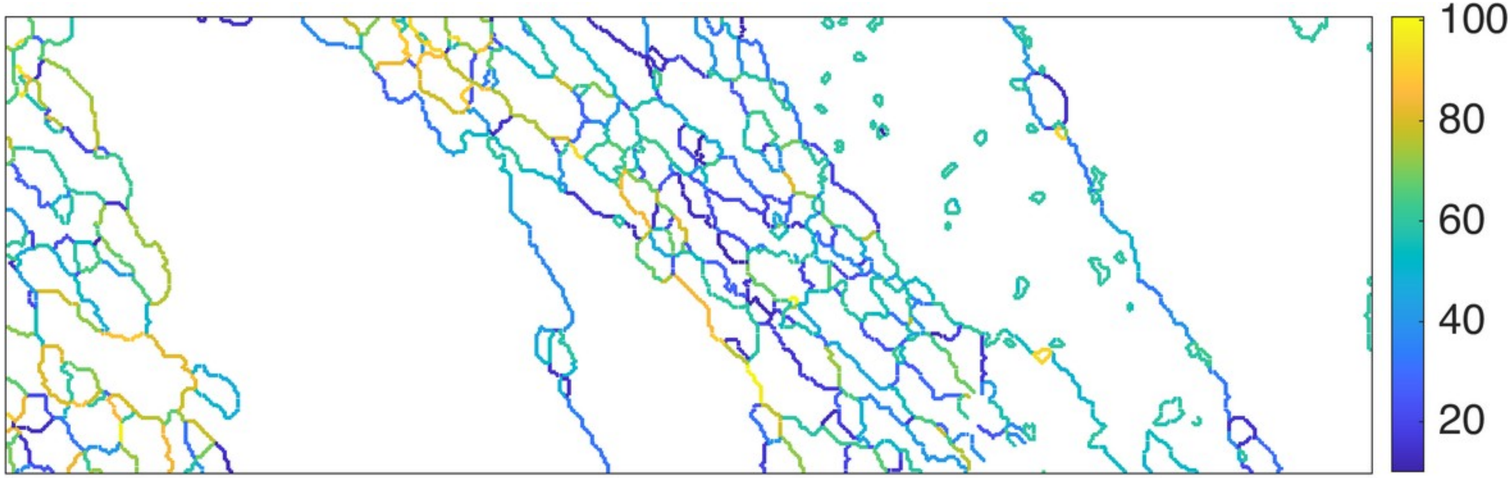


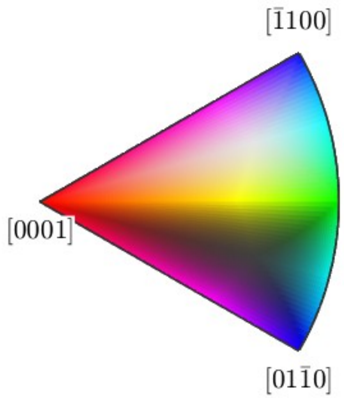
3. Grain boundaries - misorientations

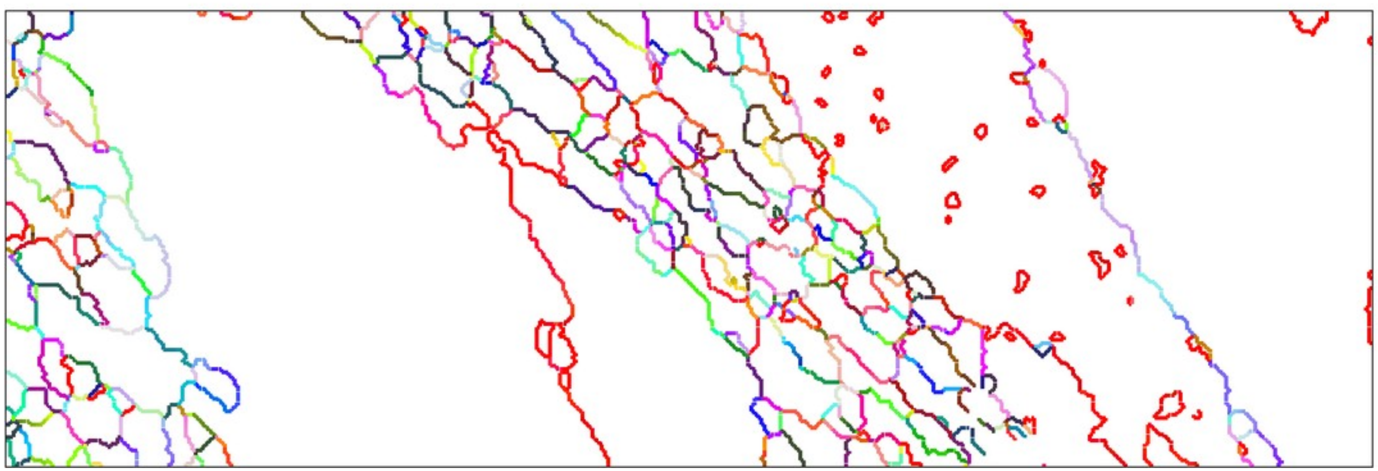
- misorientations axis-angle space

```
gbq = grains.boundary('q','q')
plot(gbq,mori.angle/degree)

ck=HSVDirectionKey(ebsd('q').CS);
plot(gbq,ck.direction2color(mori.axis))
```





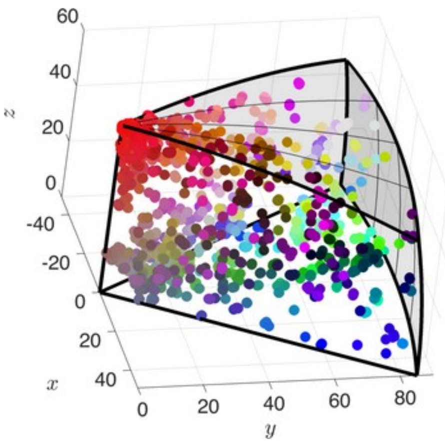


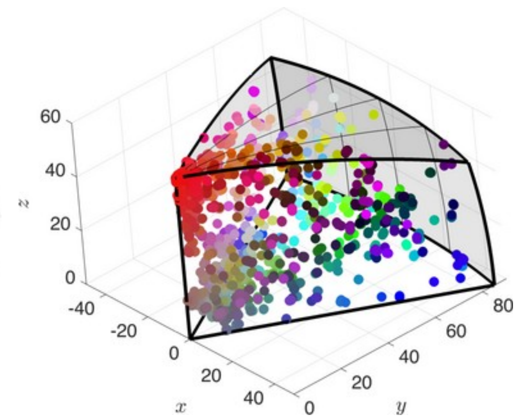


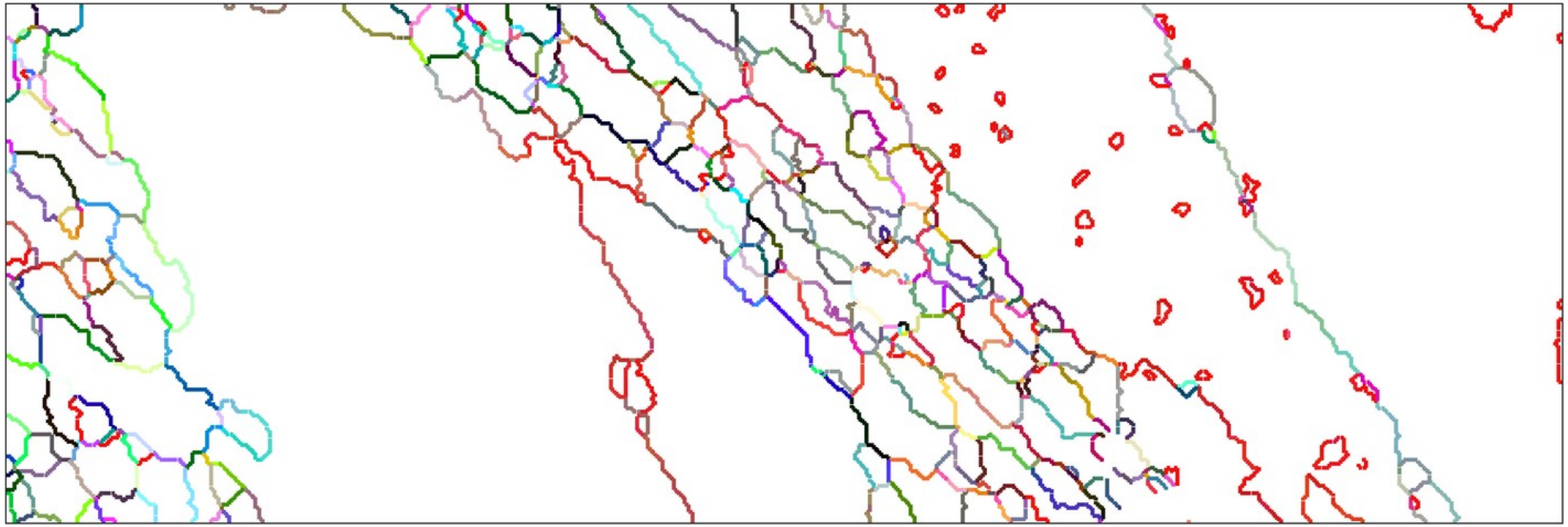
3. Grain boundaries - misorientations

- misorientations axis-angle space

```
ak = axisAngleColorKey(cs,cs,'antipodal')

plot(gbq,ak.orientation2color(mori))

plot(mori,ak.orientation2color(mori),'axisAngle')
```



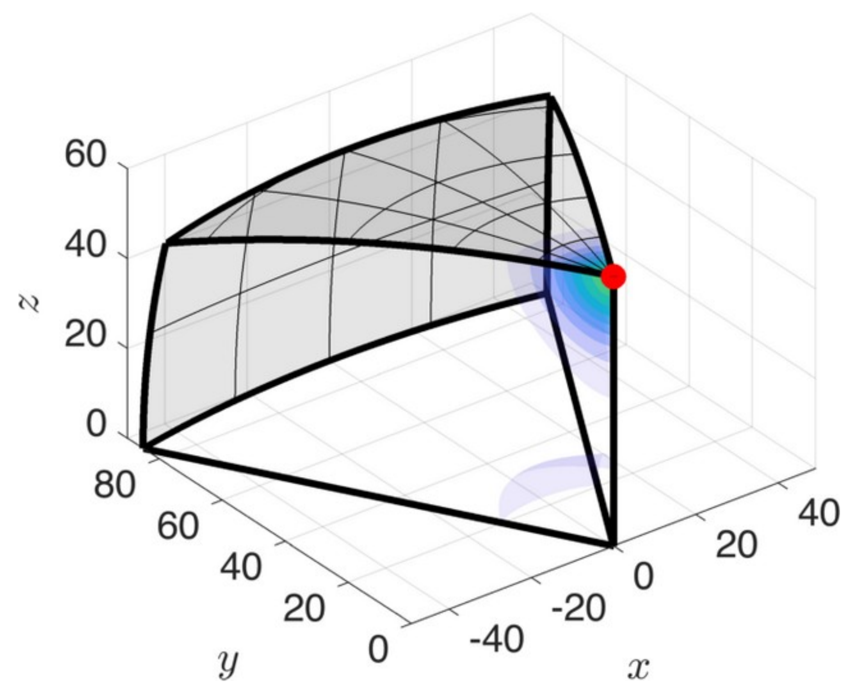




3. Grain boundaries - misorientations

- find grain boundaries with this specific misorientation e.g. a twinning relationship (here, 60° around the c-axis)

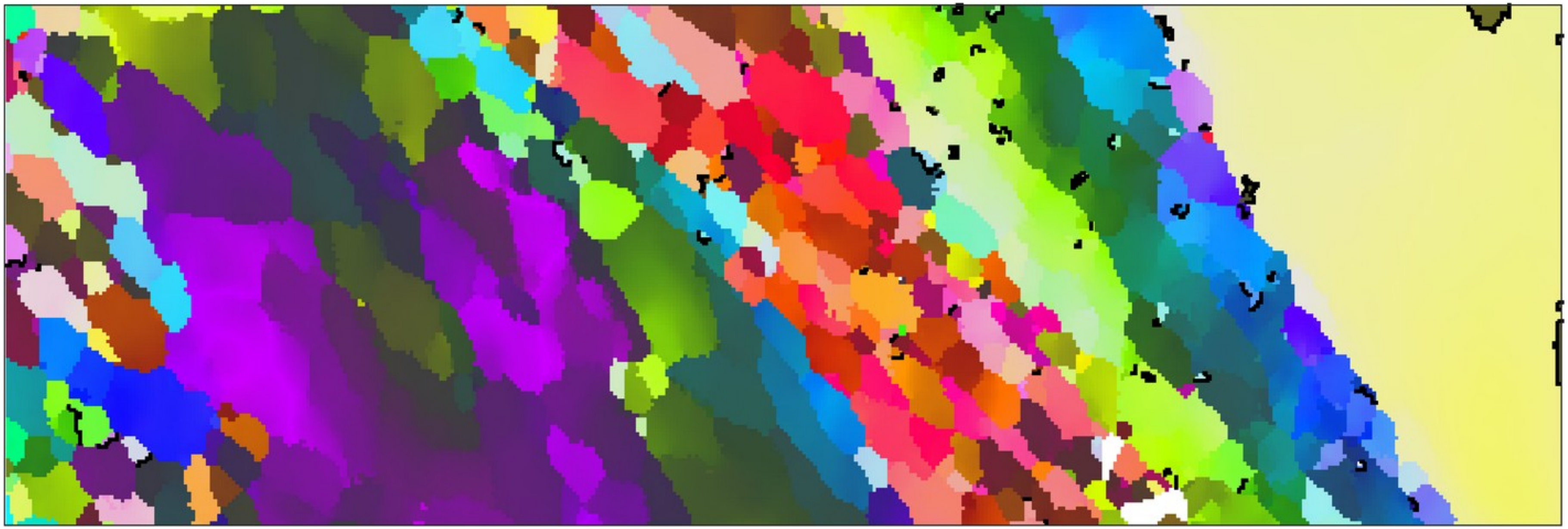
```
%define the misorientation
tR = orientation.byAxisAngle(Miller(0,0,0,1,cs),60*degree,cs,cs)
tR = misorientation (show methods, plot)
  crystal symmetry : quartz (-3m1, X||a*, Y||b, Z||c*)
  crystal symmetry : quartz (-3m1, X||a*, Y||b, Z||c*)

  Bunge Euler angles in degree
  phi1  Phi phi2 Inv.
    60    0    0    0
```



3. Grain boundaries - misorientations

- find grain boundaries with this specific misorientation / twinning relationship (60° around the c-axis)

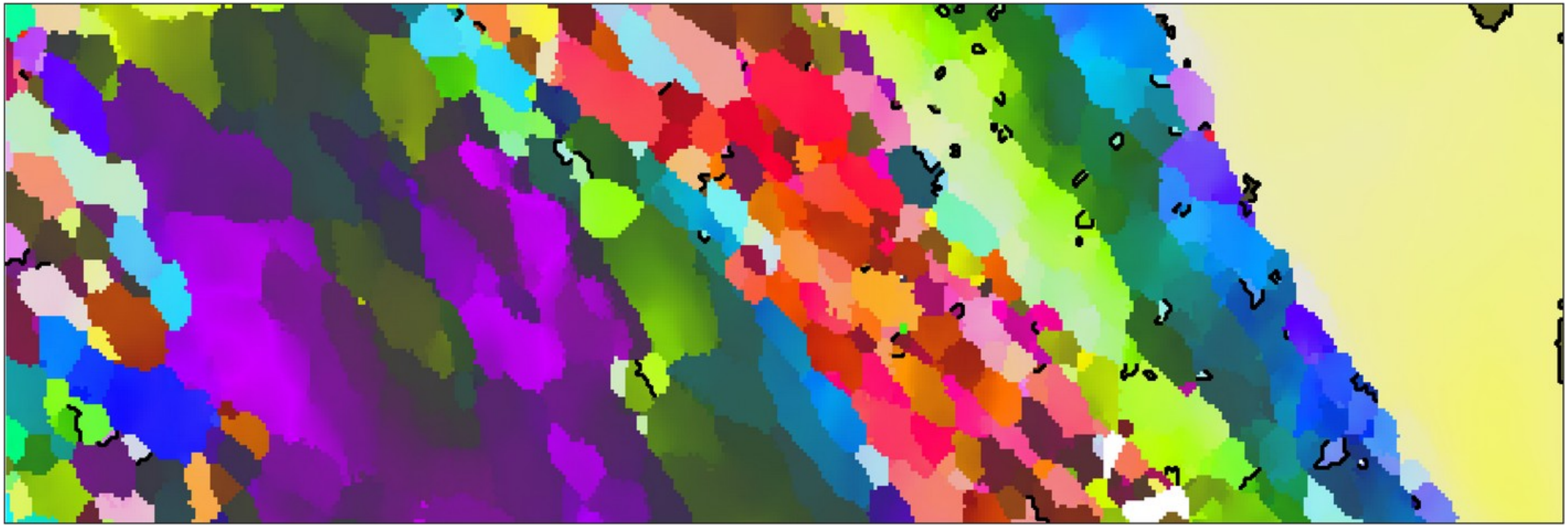
```
% decide for a threshold
istB = angle(mori,tR) < 1*degree;

ck = ipfHSVKey(cs); ck.inversePoleFigureDirection=xvector;
plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations)); hold on
plot(gbq(istB),'linewidth',1.5,'linecolor','k'); hold off
```



3. Grain boundaries - misorientations

- find grain boundaries with this specific misorientation / twinning relationship (60° around the c-axis)

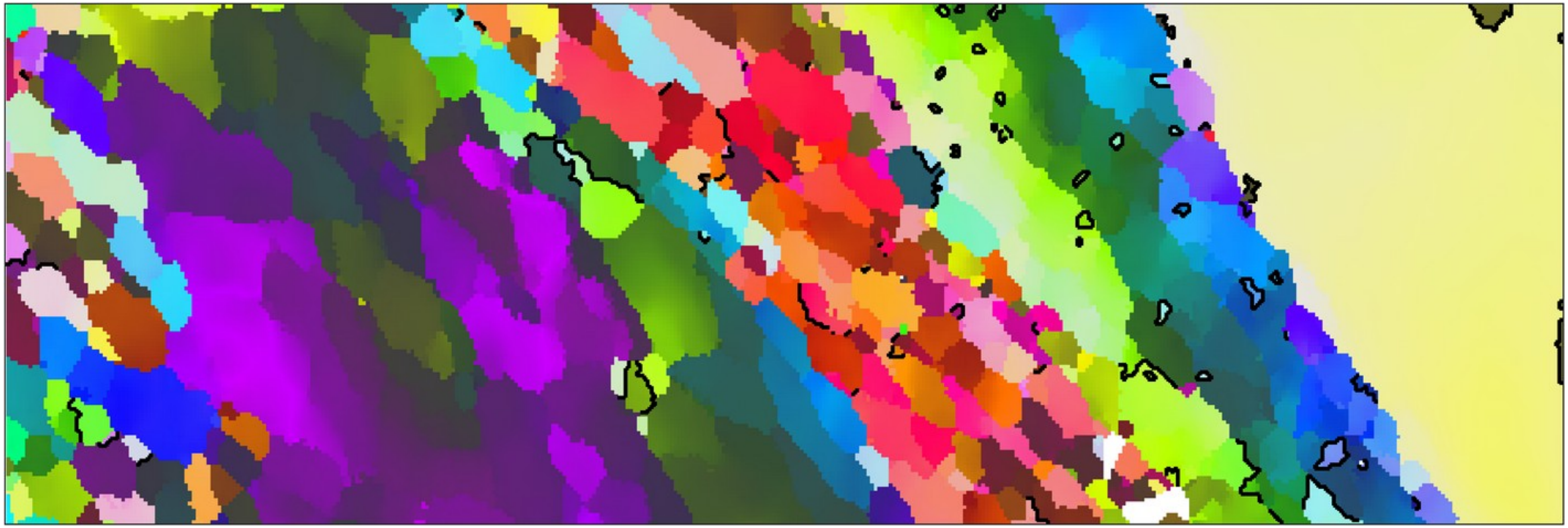
```
% decide for a threshold
istB = angle(mori,tR) < 5*degree;

ck = ipfHSVKey(cs); ck.inversePoleFigureDirection=xvector;
plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations)); hold on
plot(gbq(istB),'linewidth',1.5,'linecolor','k'); hold off
```



3. Grain boundaries - misorientations

- find grain boundaries with this specific misorientation / twinning relationship (60° around the c-axis)

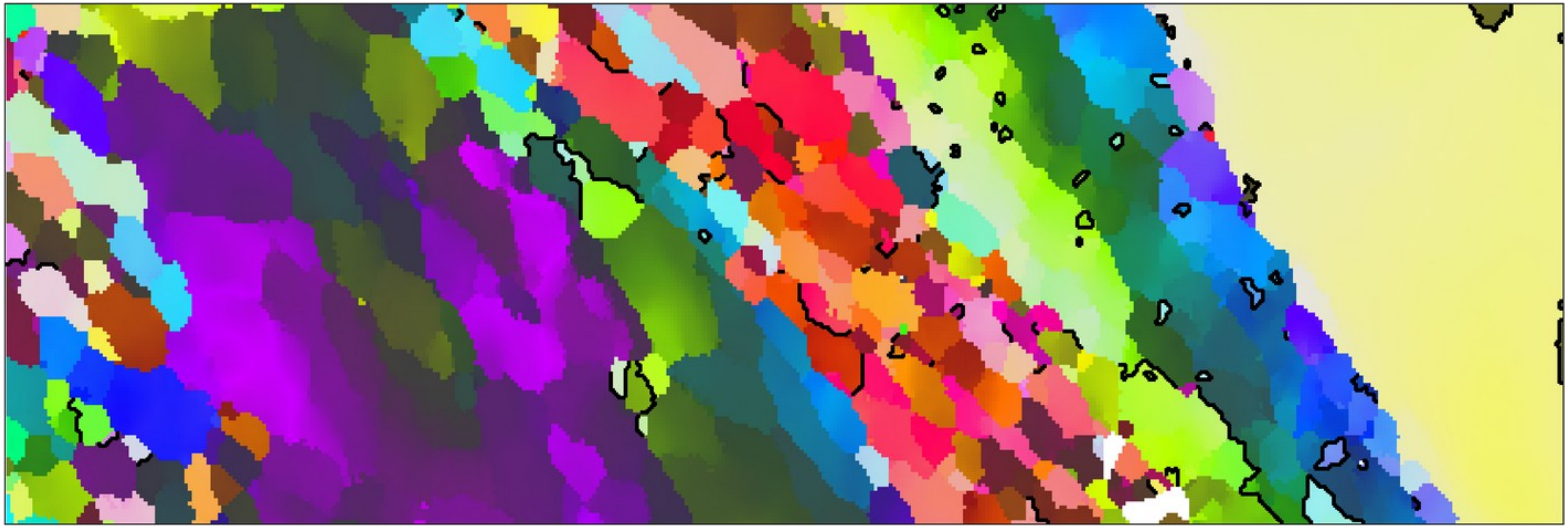
```
% decide for a threshold
istB = angle(mori,tR) < 10*degree;

ck = ipfHSVKey(cs); ck.inversePoleFigureDirection=xvector;
plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations)); hold on
plot(gbq(istB),'linewidth',1.5,'linecolor','k'); hold off
```



3. Grain boundaries - misorientations

- find grain boundaries with this specific misorientation / twinning relationship (60° around the c-axis)

```
% decide for a threshold
istB = angle(mori,tR) < 15*degree;

ck = ipfHSVKey(cs); ck.inversePoleFigureDirection=xvector;
plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations)); hold on
plot(gbq(istB),'linewidth',1.5,'linecolor','k'); hold off
```

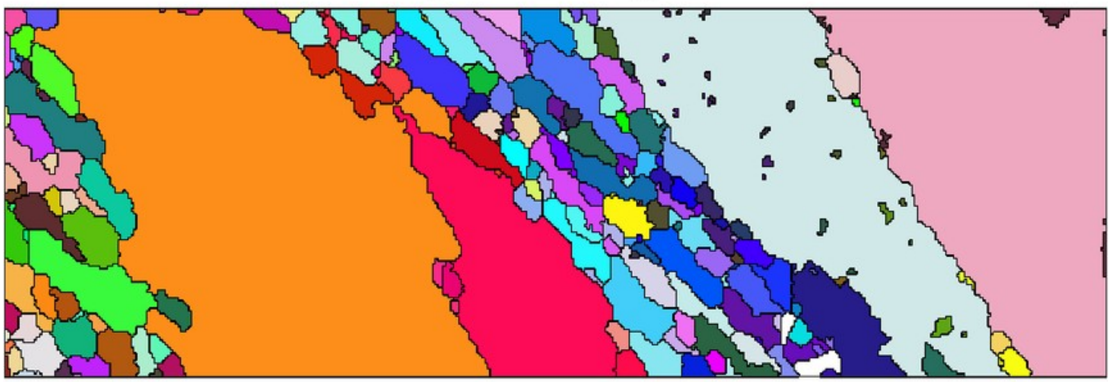


3. Grain boundaries - merging

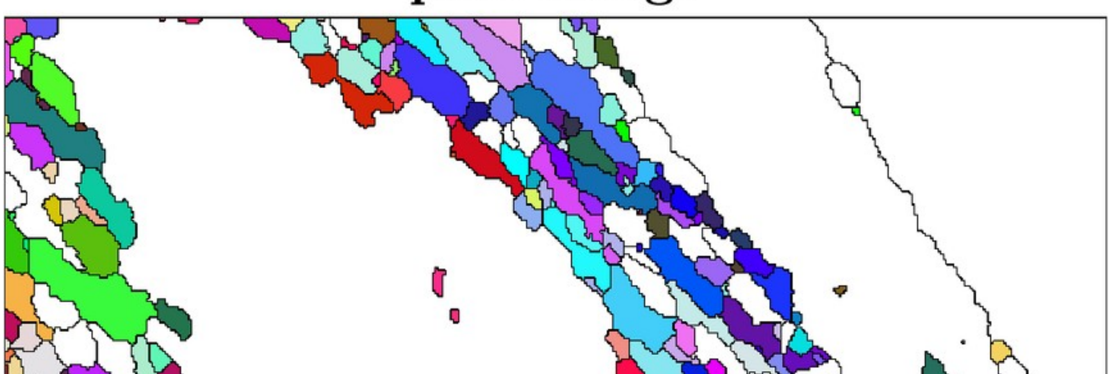
- merge twin boundaries

```
gbq= grains.boundary('q','q');
mori = gbq.misorientation;
tR = orientation.byAxisAngle(Miller(0,0,0,1,cs),60*degree,cs,cs)
tB = gbq(angle(mori,tR) < 5*degree)

[grains_m,parentId] = merge(grains,tB)
```

pre-merge



post-merge

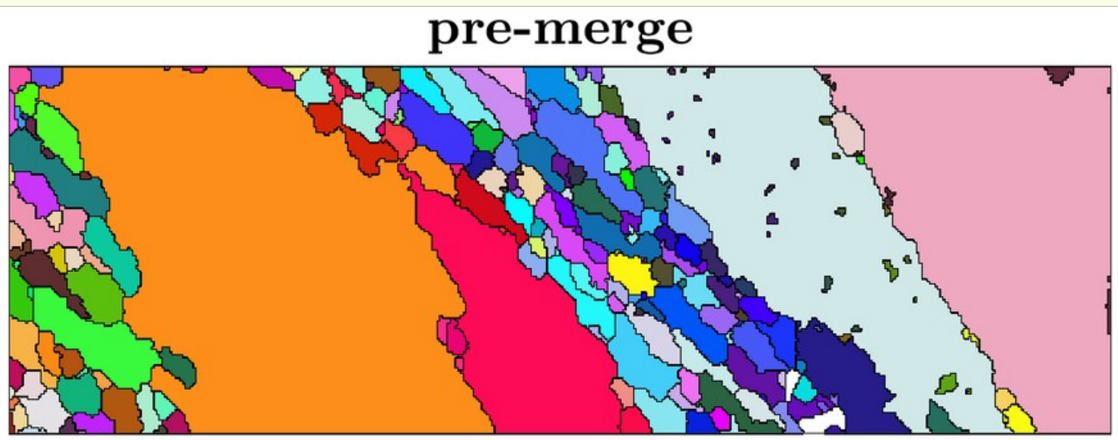


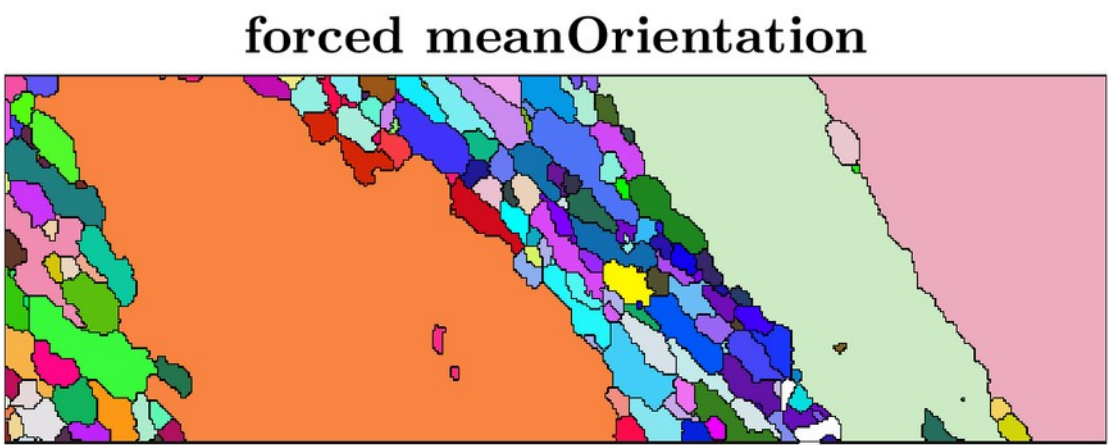
```
% update grainIds after merging
ebsd_merged = ebsd;

ebsd_merged('indexed').grainId=...
parentId(grains.id2ind(ebsd('indexed').grainId))
```

3. Grain boundaries - merging

- merge twin boundaries

```
gbq= grains.boundary('q','q');
mori = gbq.misorientation;
tR = orientation.byAxisAngle(Miller(0,0,0,1,cs),60*degree,cs,cs)
tB = gbq(angle(mori,tR) < 5*degree)

[grains_m,parentId] = merge(grains,tB,'calcMeanOrientation')
```



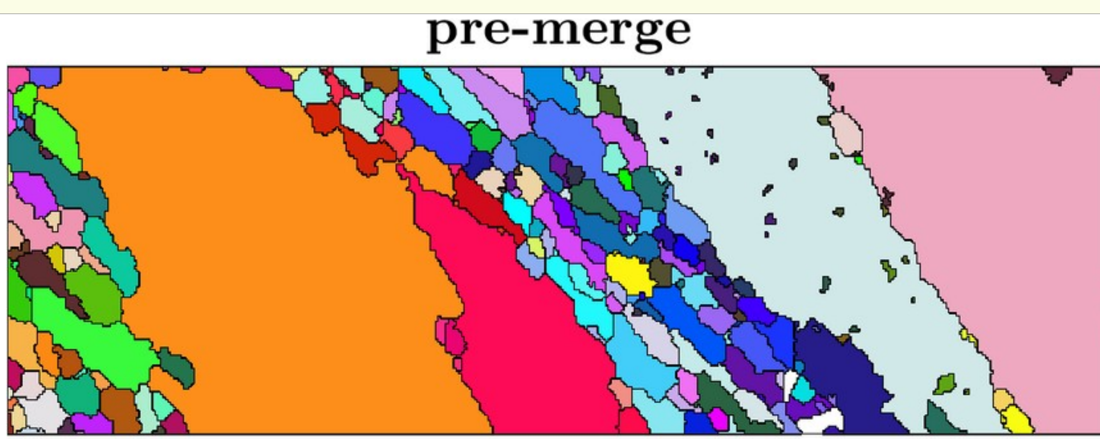


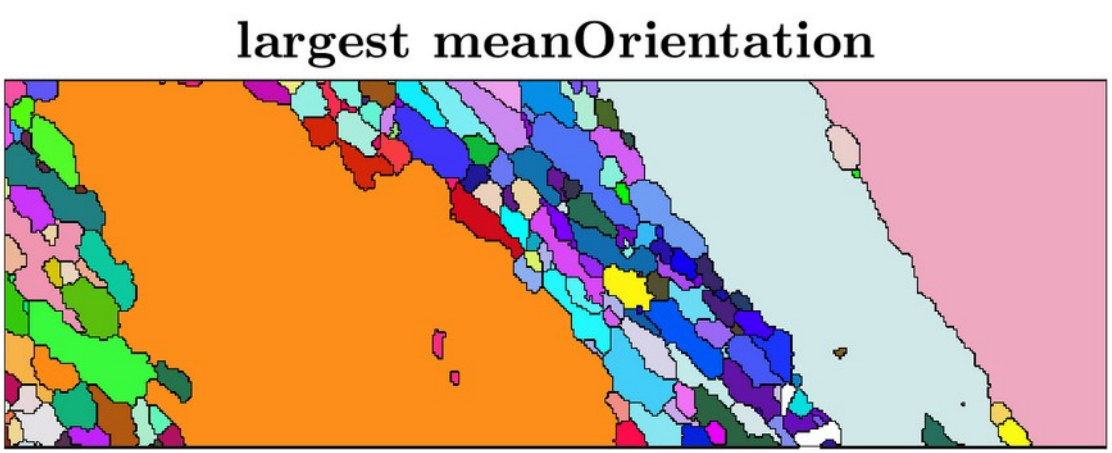
usually NOT a good idea!

3. Grain boundaries - merging

- merge twin boundaries

```
gbq= grains.boundary('q','q');
mori = gbq.misorientation;
tR = orientation.byAxisAngle(Miller(0,0,0,1,cs),60*degree,cs,cs)
tB = gbq(angle(mori,tR) < 5*degree)

[grains_m,parentId] = merge(grains,tB,'calcMeanOrientation',@largestChild)
```



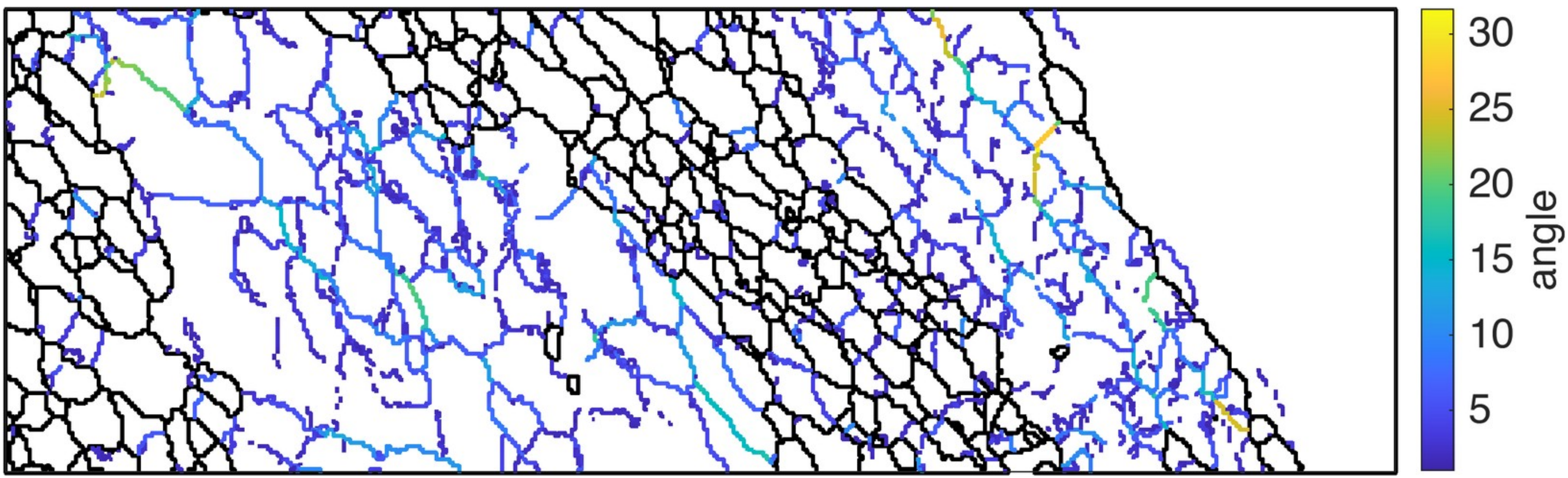



```
function mO = largestChild(grains)
 gs = grains.grainSize;
   mO =  grains(gs == max(gs)).meanOrientation;
end
```

3. Grain boundaries

- (low angle) boundaries: innerBoundary

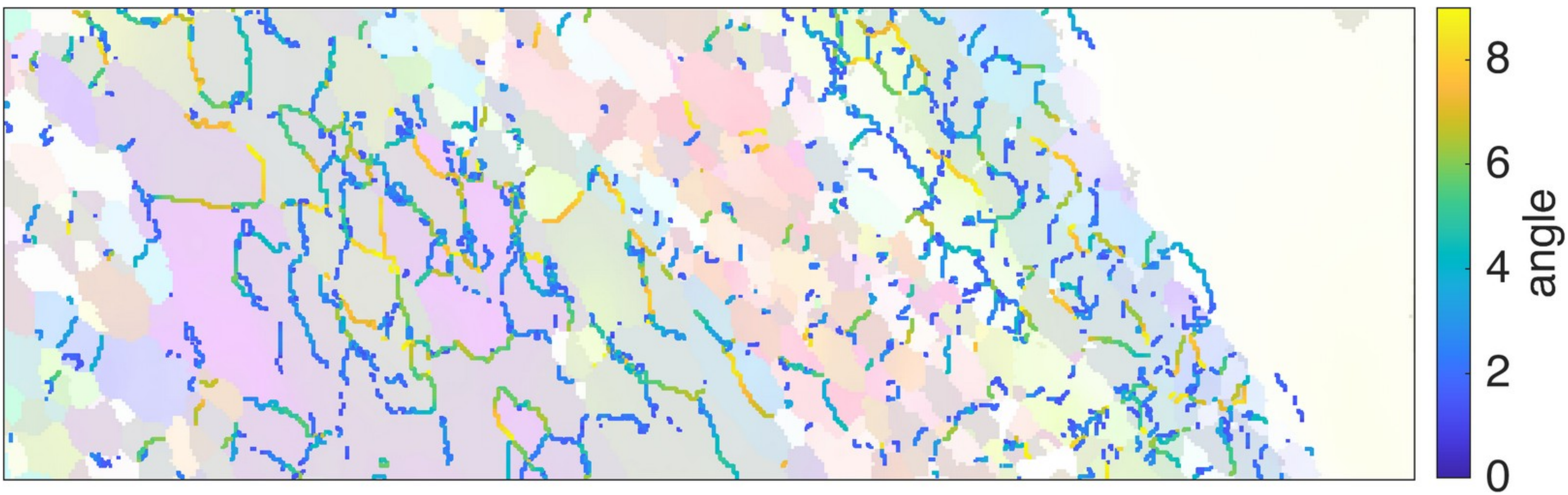
```
[grains ebsd.grainId ebsd.mis2mean] = ebsd.calcGrains('angle',[10 1]*degree)

>> min(grains.innerBoundary.misorientation.angle)/degree
 ans =
    1.0009
>> max(grains.innerBoundary.misorientation.angle)/degree
 ans =
    31.6656
```



3. Grain boundaries

- (low angle) boundaries: misorientation angle

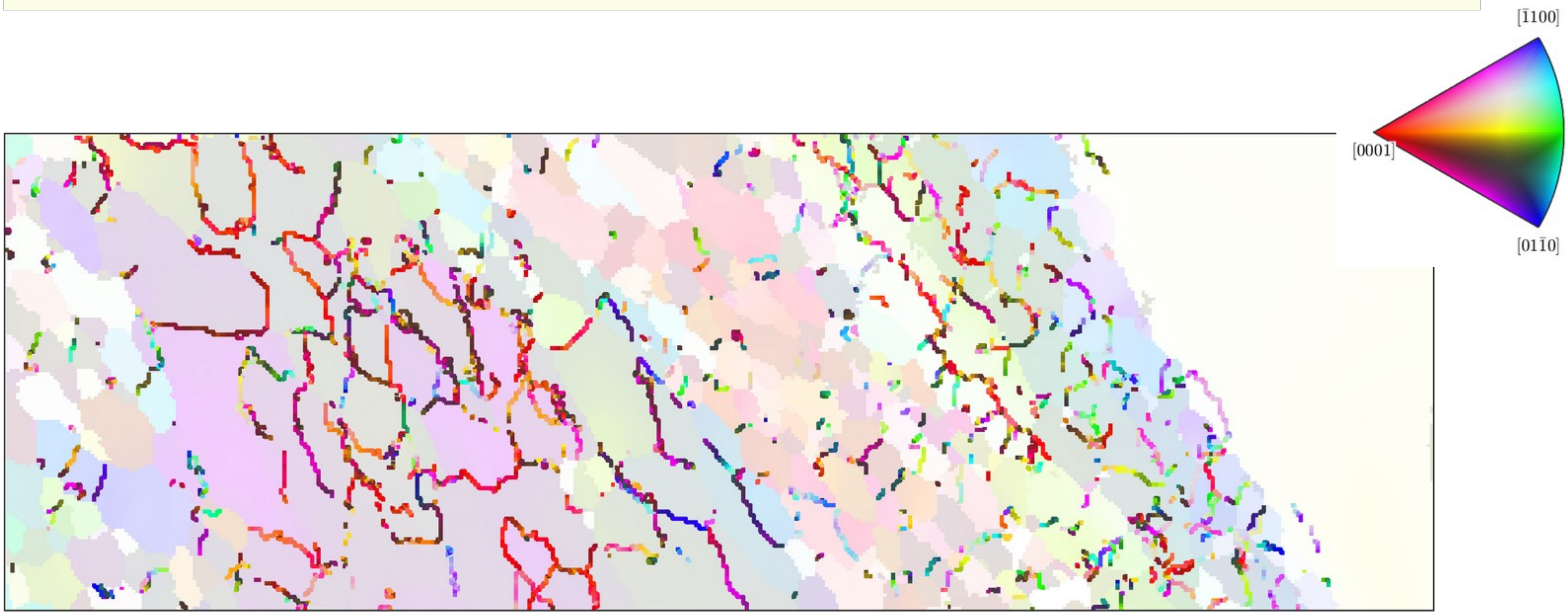
```
mori = grains.innerBoundary.misorientation;
cond = mori.angle < 9*degree;
LAB = grains('q').innerBoundary(cond);

plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations),'FaceAlpha',0.2)
hold on
plot(LAB,mori(cond).angle/degree)
hold off
```



3. Grain boundaries

- (low angle) boundaries: misorientation axes

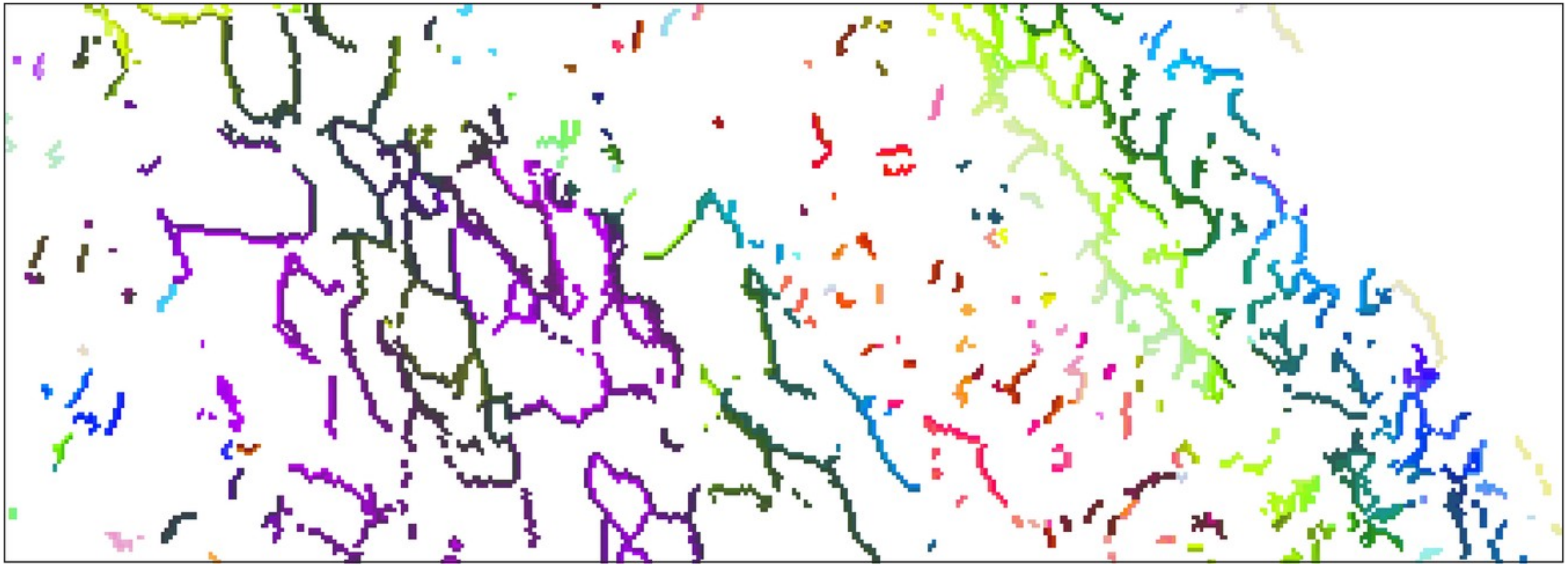
```
mori = grains.innerBoundary.misorientation;
cond = mori.angle < 9*degree;
LAB = grains('q').innerBoundary(cond);

dk = HSVDirectionKey(cs);
...
plot(LAB,dk.direction2color(mori(cond).axis))
```



3. Grain boundaries

- (low angle) boundaries: misorientation axes (specimen coordinates)

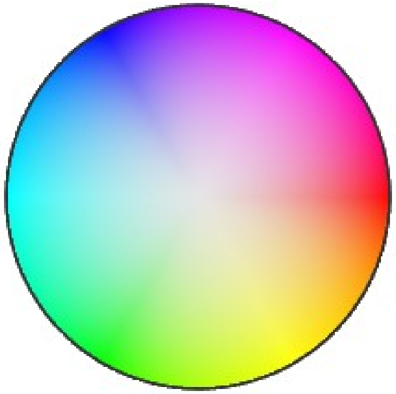
```
ebsdLAB = ebsd('id',LAB.ebsdId)

 Phase  Orientations  Mineral  Color  Symmetry  Crystal reference frame
     1  10794 (100%)   quartz   Plum      -3m1       X||a*, Y||b, Z||c*

 Properties: bands, bc, bs, error, mad, reliabilityindex, x, y, grainId, mis2mean
 Scan unit : um
 Grid size (square): 5397 x 2
```

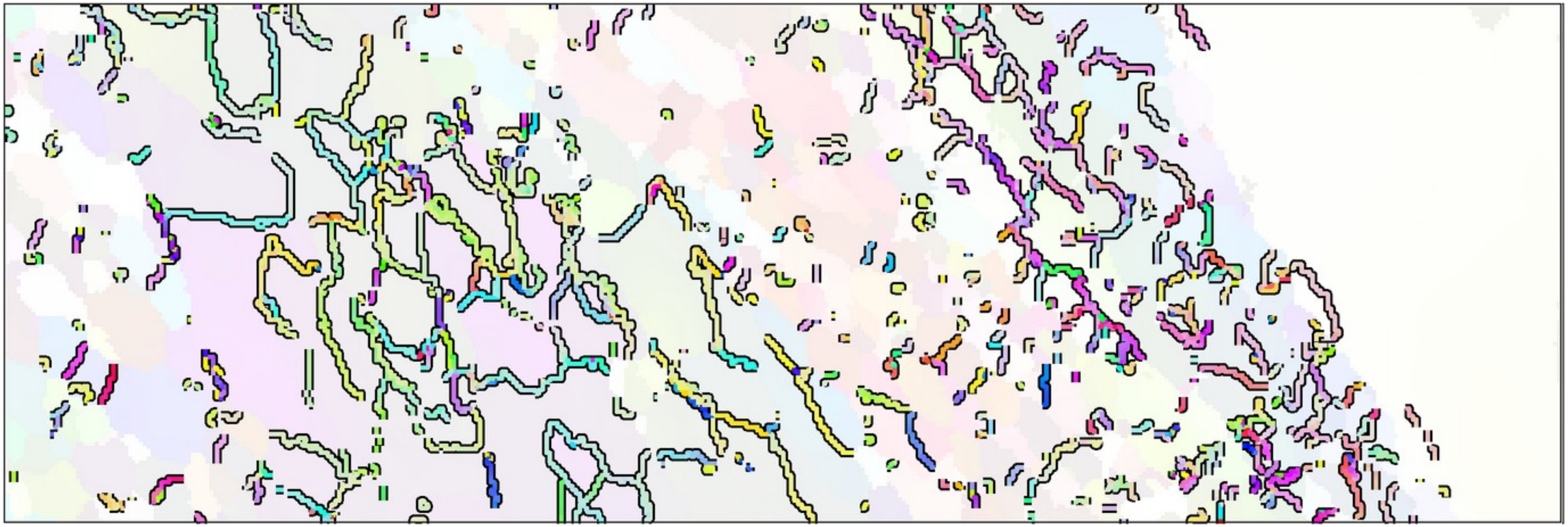


3. Grain boundaries

- (low angle) boundaries: misorientation axes (specimen coordinates)

```
o = ebsd('id',LAB.ebsdId).orientations;
o = orientation
  size: 5397 x 2
  crystal symmetry : quartz (-3m1, X||a*, Y||b, Z||c*)

axS = axis(o(:,1),o(:,2),'antipodal');

...
hold on
sk = HSVDirectionKey(specimenSymmetry('-1'))
plot(LAB,sk.direction2color(axS),'linewidth',1.5)
hold off
```

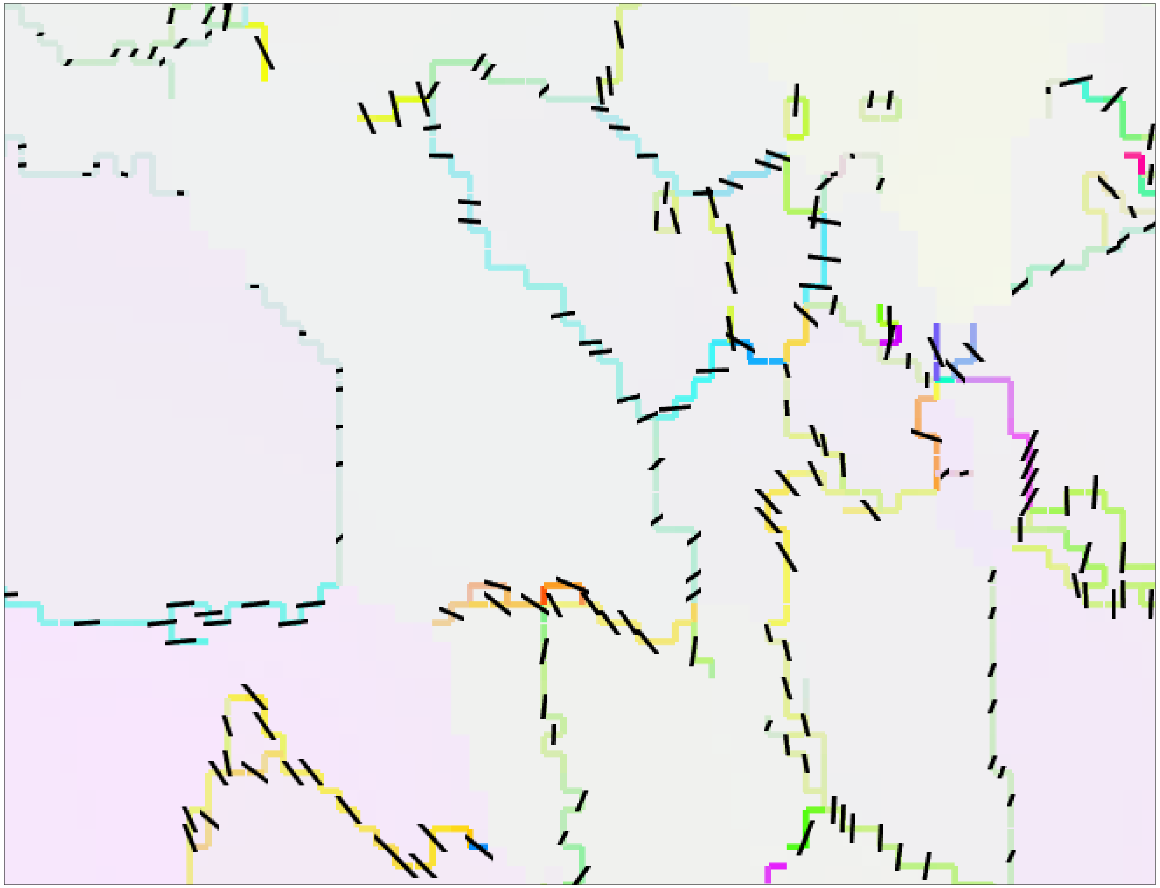


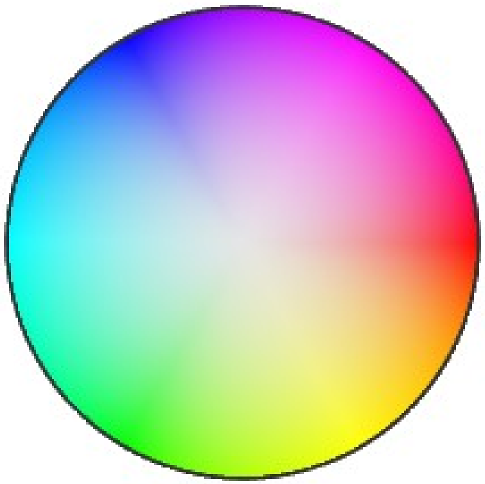


3. Grain boundaries

- (low angle) boundaries: misorientation axes (specimen coordinates)

```
plot(LAB,sk.direction2color(axS),'linewidth',1.5)
hold on
quiver(LAB(1:3:end),axS(1:3:end),'color','k')
hold off
```



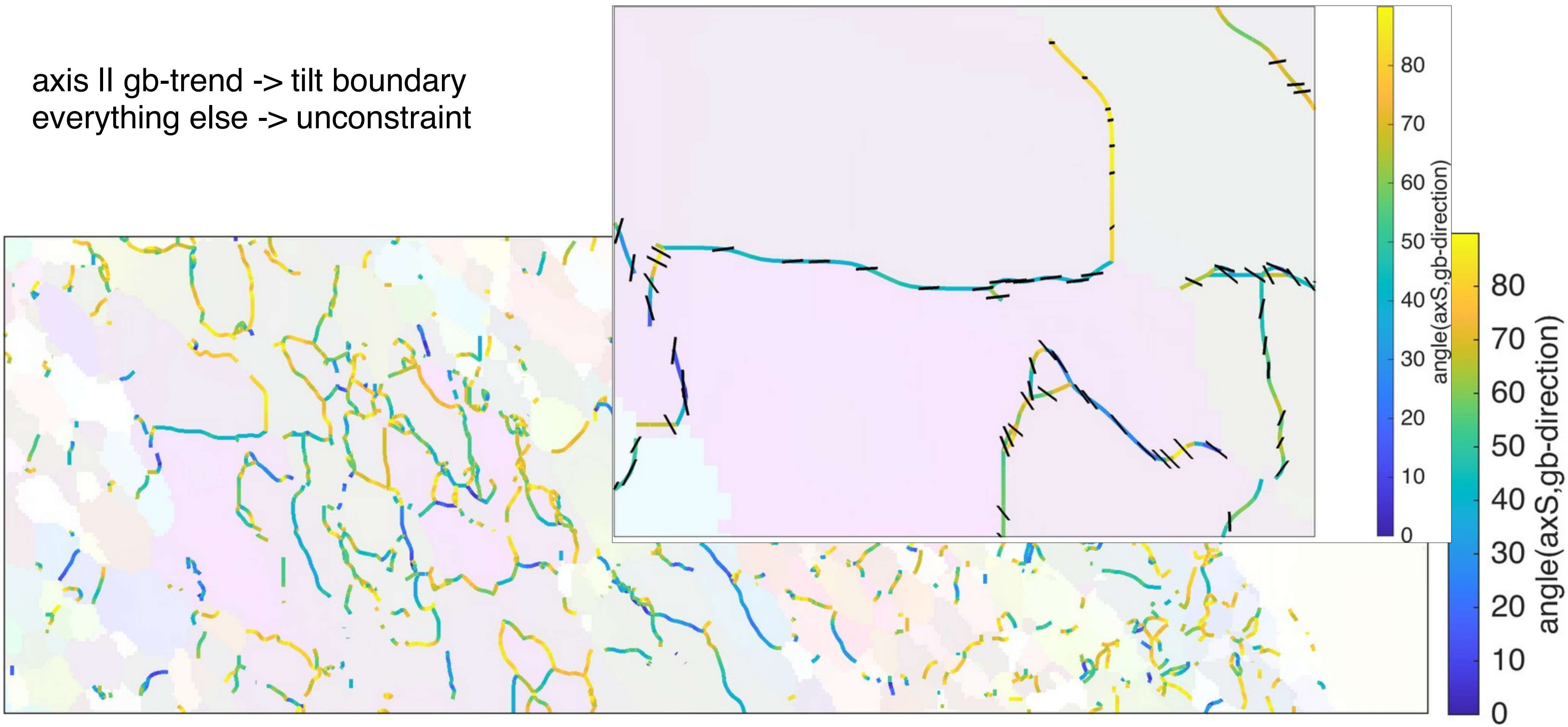


3. Grain boundaries

- (low angle) boundaries: relation between gb-direction and misorientation axis (I)

```
% angle between LAB and grain boundary direction
% tilt boundary for low angles

plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations),'FaceAlpha',0.1); hold on
plot(LAB,angle(axS,LAB.direction)/degree,'linewidth',1.5); hold off
```

axis ll gb-trend -> tilt boundary
everything else -> unconstraint

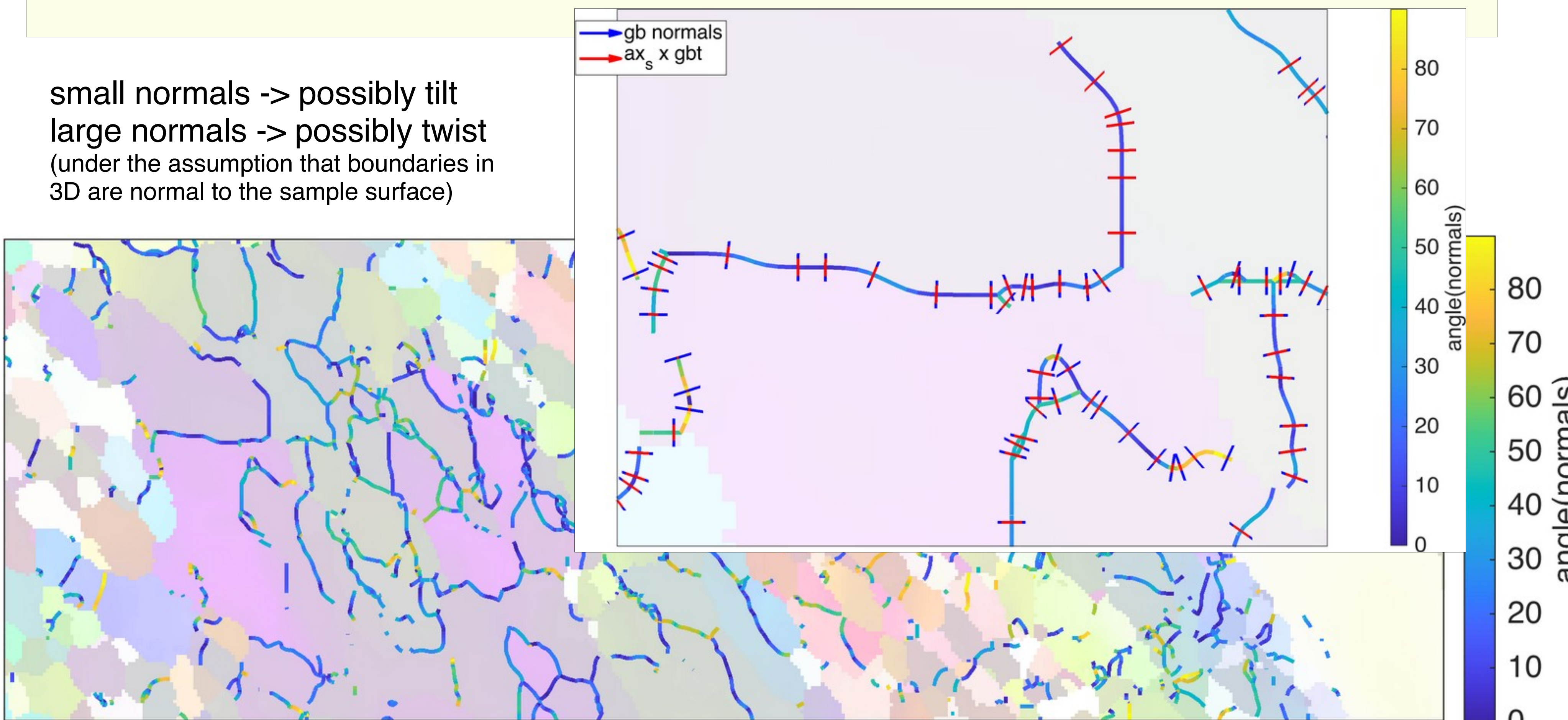


3. Grain boundaries

- (low angle) boundaries: relation between gb-direction and misorientation axis (II)

```
% angle between cross(grain boundary direction,axS) and gb-normal
gbnormal = cross(zvector,LAB.direction)
axp = cross(axS,LAB.direction)

plot(ebsd('q'),ck.orientation2color(ebsd('q').orientations),'FaceAlpha',0.2); hold on
plot(LAB,angle(gbnormal,axp)/degree,'linewidth',1.5); hold off
```

small normals -> possibly tilt
large normals -> possibly twist
(under the assumption that boundaries in 3D are normal to the sample surface)



3. Grain boundaries

- (low angle) boundaries: rotation of a single crystal direction

```
% tilt angle of c-axis
cdir = o.*Miller(0,0,0,1,cs)
cdir.antipodal=1

...
hold on
plot(LAB,angle(cdir(:,1),cdir(:,2))/degree,'linewidth',1.5)
hold off
```